Topology I

Lecture notes, Chapters 1–8 (General Topology)

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Chapter 1. Set Theory and Logic

Bracketed numbers after a result refer to Munkres.

1.1 Fundamental concepts

Motivation. Topology uses sets and functions heavily; this chapter fixes the language and the facts (countability, choice, well-ordering) that later proofs rely on.

We use sets, membership, inclusion $A \subset B$, union, intersection, difference $A - B$, the empty set, and the cartesian product $A \times B$ of ordered pairs. For a collection $\mathcal{A}$ of sets, $\bigcup_{A\in\mathcal{A}} A$ and $\bigcap_{A\in\mathcal{A}} A$ are defined as usual (the intersection of the empty collection is left undefined). De Morgan's laws hold:

$$A - \bigcup_\alpha B_\alpha = \bigcap_\alpha (A - B_\alpha), \qquad A - \bigcap_\alpha B_\alpha = \bigcup_\alpha (A - B_\alpha),$$

and $A \cap \bigcup B_\alpha = \bigcup (A \cap B_\alpha)$, $A \cup \bigcap B_\alpha = \bigcap (A \cup B_\alpha)$. In logic, "$P \Rightarrow Q$" is equivalent to its contrapositive "not $Q \Rightarrow$ not P" but not to its converse; the negation of "for every x, $P(x)$" is "for some x, not $P(x)$".

1.2 Functions

A **function** $f : A \to B$ is a rule (formally a subset of $A\times B$ in which each a appears as first coordinate of exactly one pair). For $A_0 \subset A$, $f(A_0)$ is the image; for $B_0 \subset B$, $f^{-1}(B_0) = \{a : f(a) \in B_0\}$. f is **injective** if $f(a) = f(a')$ implies $a = a'$, **surjective** if $f(A) = B$, **bijective** if both; a bijection has an inverse $f^{-1} : B \to A$. Composites of injections (surjections) are injections (surjections). Preimages preserve unions, intersections and differences; images preserve unions and inclusions only, and $A_0 \subset f^{-1}(f(A_0))$, $f(f^{-1}(B_0)) \subset B_0$, with equality if f is injective (resp. surjective).

Lemma 1.1 (2.1)

Let $f : A \to B$. If there are $g, h : B \to A$ with $g(f(a)) = a$ for all $a \in A$ and $f(h(b)) = b$ for all $b \in B$, then f is bijective and $g = h = f^{-1}$.

Proof. $f(a) = f(a')$ gives $a = g(f(a)) = g(f(a')) = a'$, so f is injective; $b = f(h(b))$ shows it is surjective. Then $g = g \circ (f \circ h) = (g \circ f) \circ h = h$, and both equal f^{-1}. ∎

1.3 Relations

Motivation. Equivalence relations will give quotient spaces and order relations the order topology, both introduced here set-theoretically.

A **relation** on A is a subset $C \subset A \times A$, written xCy. An **equivalence relation** $\sim$ is reflexive, symmetric and transitive; the **equivalence class** of x is $E = \{y : y \sim x\}$.

Lemma 1.2 (3.1)

Two equivalence classes E, E' are either disjoint or equal.

Proof. If $y \in E \cap E'$ with E the class of x and E' the class of x', then $y \sim x$ and $y \sim x'$, so $x \sim x'$. Any $w \sim x$ then satisfies $w \sim x'$; so $E \subset E'$ and symmetrically. ∎

Hence the classes form a **partition** of A (disjoint nonempty sets with union A); conversely every partition comes from the relation "lying in the same piece".

An **order relation** (simple or linear order) $<$ satisfies: comparability ($x \neq y \Rightarrow x < y$ or $y < x$), nonreflexivity (no $x < x$) and transitivity. In an ordered set we have open intervals (a, b), immediate successors and predecessors, and the **dictionary order** on $A \times B$: $a_1 \times b_1 < a_2 \times b_2$ iff $a_1 < a_2$, or $a_1 = a_2$ and $b_1 < b_2$. Ordered sets A, B have the same **order type** if there is an order-preserving bijection. For $A_0 \subset A$, an **upper bound** b satisfies $x \leq b$ for all $x \in A_0$; the least one is the **supremum** $\sup A_0$, and dually $\inf A_0$. A has the **least upper bound property** if every nonempty subset bounded above has a supremum; this is equivalent to the greatest lower bound property.

1.4 The integers and the real numbers

Motivation. We record the axioms of $\mathbb{R}$, chiefly the least upper bound property, and induction on $\mathbb{Z}_+$, so later proofs can cite exactly what they use.

We assume $\mathbb{R}$ is an ordered field with the least upper bound property; $\mathbb{Z}_+$ is the smallest inductive subset (containing 1 and closed under $x \mapsto x + 1$), which gives the principle of induction.

Theorem 1.3 (4.1, well-ordering property)

Every nonempty subset of $\mathbb{Z}_+$ has a smallest element.

Proof. First, for each n, every nonempty subset of $\{1, \ldots, n\}$ has a smallest element: the set A of n for which this holds contains 1, and if $n \in A$ and $C \subset \{1, \ldots, n+1\}$, then either $C = \{n+1\}$ or $C \cap \{1, \ldots, n\}$ has a least element, which is the least element of C. For a nonempty $D \subset \mathbb{Z}_+$ choose $n \in D$; the least element of $D \cap \{1, \ldots, n\}$ is the least element of D. ■

Theorem 1.4 (4.2, strong induction)

Let $A \subset \mathbb{Z}_+$ be such that $\{1, \ldots, n-1\} \subset A$ implies $n \in A$, for each n. Then $A = \mathbb{Z}_+$.

Proof. Otherwise let n be the least element of $\mathbb{Z}_+ - A$; then $\{1, \ldots, n-1\} \subset A$, so $n \in A$, a contradiction. ■

The **Archimedean ordering property** ($\mathbb{Z}_+$ is unbounded above in $\mathbb{R}$) follows from the least upper bound property: if $b = \sup \mathbb{Z}_+$ existed, $b - 1 < n$ for some n, so $n + 1 > b$.

1.5 Cartesian products

For an indexed family $\{A_\alpha\}_{\alpha \in J}$, the **cartesian product** $\prod_{\alpha \in J} A_\alpha$ is the set of all functions $x : J \to \bigcup A_\alpha$ with $x(\alpha) \in A_\alpha$ (written $(x_\alpha)_{\alpha \in J}$). For finite $J = \{1, \ldots, m\}$ these are m-tuples; for $J = \mathbb{Z}_+$ they are sequences; X^J denotes the set of all functions $J \to X$, e.g. $\mathbb{R}^\omega$, the set of real sequences.

1.6 Finite sets

Motivation. Finiteness is defined through bijections with $\{1, \ldots, n\}$, and its intuitive properties, including the pigeonhole principle, are proved.

A set is **finite** if it is empty or in bijection with a section $\{1, \ldots, n\}$; we show n is unique.

Lemma 1.5 (6.1)

Let $a_0 \in A$. There is a bijection $A \to \{1, \dots, n+1\}$ iff there is a bijection $A - \{a_0\} \to \{1, \dots, n\}$.

Proof. Given $g : A - \{a_0\} \to \{1, \dots, n\}$, extend by $a_0 \mapsto n+1$. Conversely, given $f : A \to \{1, \dots, n+1\}$, if $f(a_0) = n+1$ restrict; otherwise compose f with the transposition of $f(a_0)$ and $n+1$ first. ■

Theorem 1.6 (6.2)

Let $f : A \to \{1, \dots, n\}$ be a bijection and B a proper subset of A. Then there is no bijection $B \to \{1, \dots, n\}$, but (if $B \neq \emptyset$) there is a bijection $B \to \{1, \dots, m\}$ for some $m < n$.

Proof. Induction on n. For $n = 1$, A is a point and $B = \emptyset$. Assume the result for n and let $f : A \to \{1, \dots, n+1\}$, $B \subsetneq A$ nonempty. Pick $a_0 \in B$; $A - \{a_0\}$ is in bijection with $\{1, \dots, n\}$ by Lemma 1.5. Since $B - \{a_0\} \subsetneq A - \{a_0\}$, the induction hypothesis says $B - \{a_0\}$ is empty or in bijection with $\{1, \dots, p\}$, $p < n$, and not with $\{1, \dots, n\}$; by Lemma 1.5 again, B is in bijection with $\{1, \dots, p+1\}$, $p+1 < n+1$, and not with $\{1, \dots, n+1\}$. ■

Corollary 1.7 (6.3–6.5)

A finite set has no bijection with a proper subset of itself; $\mathbb{Z}_+$ is not finite; the cardinality of a finite set is uniquely determined.

Proof. The first is Theorem 1.6. $n \mapsto n+1$ is a bijection of $\mathbb{Z}_+$ onto a proper subset. If A were in bijection with $\{1, \dots, m\}$ and $\{1, \dots, n\}$, $m < n$, then $\{1, \dots, n\}$ would be in bijection with its proper subset $\{1, \dots, m\}$. ■

Corollary 1.8 (6.6)

A subset B of a finite set A is finite; if B is proper, its cardinality is smaller.

Corollary 1.9 (6.7)

For nonempty B the following are equivalent:
(1) B is finite;
(2) there is a surjection from some section $\{1, \dots, n\}$ onto B;
(3) there is an injection of B into some section.

Proof. (1)⇒(2) clear. (2)⇒(3): send b to the least element of $f^{-1}(b)$ (Theorem 1.3). (3)⇒(1): B is in bijection with a subset of a finite set; use Corollary 1.8. ■

Corollary 1.10 (6.8)

Finite unions and finite cartesian products of finite sets are finite.

Proof. For $A \cup B$ with $A \cong \{1, \dots, m\}$ and $B \cong \{1, \dots, n\}$, the map onto $\{1, \dots, m+n\}$ sending B to the last n values gives a surjection from a section; apply Corollary 1.9. $A \times B$ is the union of the finite sets $A \times \{b\}$; induct for more factors. ■

1.7 Countable and uncountable sets

Motivation. Countability criteria are needed for second countability and separability, and Cantor's diagonal argument shows not every infinite set is countable.

A set is **countably infinite** if it is in bijection with $\mathbb{Z}_+$, **countable** if finite or countably infinite, and **uncountable** otherwise.

Theorem 1.11 (7.1)

For nonempty B the following are equivalent:
(1) B is countable;
(2) there is a surjection $f : \mathbb{Z}_+ \to B$;
(3) there is an injection $g : B \to \mathbb{Z}_+$.

Proof. (1)⇒(2): if B is finite, extend a surjection from a section by a constant. (2)⇒(3): $g(b) = \min f^{-1}(b)$. (3)⇒(1): $g(B) \subset \mathbb{Z}_+$ is countable by the next lemma. ■

Lemma 1.12 (7.2)

An infinite subset $C \subset \mathbb{Z}_+$ is countably infinite.

Proof. Define $h(1) = \min C$ and $h(n) = \min\{C - h(\{1, \dots, n-1\})\}$, which is nonempty because C is infinite (recursive definition, Theorem 1.21). h is injective (increasing). It is surjective: if $c \in C$ were not in the image, then $h(n) < c$ for all n (c is always available), so h would map $\mathbb{Z}_+$ injectively into $\{1, \dots, c\}$, contradicting Corollary 1.7. ■

Corollary 1.13 (7.3)

A subset of a countable set is countable.

Corollary 1.14 (7.4)

$\mathbb{Z}_+ \times \mathbb{Z}_+$ is countably infinite.

Proof. $(n, m) \mapsto 2^n 3^m$ is injective by unique factorization; apply Theorem 1.11. ■

Theorem 1.15 (7.5)

A countable union of countable sets is countable.

Proof. Let $\{A_n\}_{n \in J}$, $J \subset \mathbb{Z}_+$, be countable and nonempty. Choose surjections $f_n : \mathbb{Z}_+ \to A_n$ (and, if J is finite, a surjection $g : \mathbb{Z}_+ \to J$). Then $(k, m) \mapsto f_{g(k)}(m)$ maps $\mathbb{Z}_+ \times \mathbb{Z}_+$ onto $\bigcup A_n$; compose with a surjection $\mathbb{Z}_+ \to \mathbb{Z}_+ \times \mathbb{Z}_+$. (The simultaneous choice of the f_n uses the axiom of choice.) ■

Theorem 1.16 (7.6)

A finite product of countable sets is countable.

Proof. $A \times B$ is the image of $\mathbb{Z}_+ \times \mathbb{Z}_+$ under $(n, m) \mapsto (f(n), g(m))$; induct on the number of factors. ■

Theorem 1.17 (7.7)

$\{0,1\}^\omega$ is uncountable.

Proof. Given $g : \mathbb{Z}_+ \to \{0,1\}^\omega$, write $g(n) = (x_{n1}, x_{n2}, \dots)$ and let $y_n = 1 - x_{nn}$. Then $y = (y_n)$ differs from $g(n)$ in the n-th coordinate, so g is not surjective. ■

Theorem 1.18 (7.8)

For any set A there is no injection $\mathcal{P}(A) \to A$ and no surjection $A \to \mathcal{P}(A)$.

Proof. An injection $\mathcal{P}(A) \to A$ would yield a surjection $A \to \mathcal{P}(A)$. Given $g : A \to \mathcal{P}(A)$, let $B = \{a : a \notin g(a)\}$. If $B = g(a_0)$, then $a_0 \in B \iff a_0 \notin g(a_0) = B$, a contradiction. ■

1.8 The principle of recursive definition

Motivation. The recursion principle justifies step-by-step definitions "choose x_1, then x_2, ...".

Fix a set C and the requirement $(*)$: $h(1) =$ the least element of C, and $h(i) =$ the least element of $C - h(\{1, \dots, i-1\})$ (with $C \subset \mathbb{Z}_+$ infinite, as in Lemma 1.12); the general case is identical.

Lemma 1.19 (8.1)

For each n there is a function $f : \{1, \dots, n\} \to C$ satisfying $(*)$ on its domain.

Proof. Induction: extend a function on $\{1, \dots, n\}$ by the value prescribed by $(*)$ at $n+1$ (the set $C - f(\{1, \dots, n\})$ is nonempty since C is infinite). ■

Lemma 1.20 (8.2)

Two functions on $\{1, \dots, n\}$ and $\{1, \dots, m\}$ satisfying $(*)$ agree on their common domain.

Proof. If not, let i be the least point of disagreement; $(*)$ determines the value at i from the (equal) values below i. ■

Theorem 1.21 (8.3–8.4, principle of recursive definition)

Let A be a set, $a_0 \in A$, and ρ a function assigning to each function f from a nonempty section $\{1, \dots, n\}$ into A an element $\rho(f) \in A$. Then there is a unique $h : \mathbb{Z}_+ \to A$ with $h(1) = a_0$ and $h(i) = \rho(h|_{\{1,\dots,i-1\}})$ for $i > 1$.

Proof. As in Lemmas 1.19 and 1.20, for each n there is a unique function on $\{1, \dots, n\}$ satisfying the formula, and these are compatible; their union is a function h on $\mathbb{Z}_+$ with the property, unique by the same argument. ■

1.9 Infinite sets and the axiom of choice

Motivation. The axiom of choice legitimizes infinitely many arbitrary choices, for instance a countably infinite subset of every infinite set.

Theorem 1.22 (9.1)

For a set A the following are equivalent:
(1) there is an injection $\mathbb{Z}_+ \to A$;
(2) there is a bijection of A with a proper subset of itself;
(3) A is infinite.

Proof. (1)⇒(2): if $f : \mathbb{Z}_+ \to A$ is injective, define $g(f(n)) = f(n+1)$ and $g(x) = x$ off $f(\mathbb{Z}_+)$; g is a bijection onto $A - \{f(1)\}$.

(2)⇒(3) is Corollary 1.7. (3)⇒(1): let c be a choice function on the nonempty subsets of A (Lemma 1.23) and define recursively $f(1) = c(A)$, $f(n) = c(A - f(\{1, \dots, n-1\}))$; the sets are nonempty because A is infinite, and f is injective. ■

Definition (Axiom of choice)

Given a collection $\mathcal{A}$ of disjoint nonempty sets, there is a set C consisting of exactly one element from each member of $\mathcal{A}$.

Lemma 1.23 (9.2, choice function)

Given a collection $\mathcal{B}$ of nonempty sets (not necessarily disjoint), there is a function $c : \mathcal{B} \to \bigcup \mathcal{B}$ with $c(B) \in B$ for each B.

Proof. The sets $B' = \{B\} \times B$, $B \in \mathcal{B}$, are disjoint and nonempty. A set C meeting each B' in one point is a set of pairs (B, x) with exactly one pair for each B; it is the desired function. ■

1.10 Well-ordered sets

Motivation. Well-ordered sets allow induction beyond $\mathbb{Z}_+$, and the minimal uncountable well-ordered set S_Ω is a source of counterexamples.

An ordered set is **well-ordered** if every nonempty subset has a smallest element. Subsets of well-ordered sets are well-ordered, and so are dictionary-ordered finite products of well-ordered sets. For α in a well-ordered set X, the **section** is $S_\alpha = \{x : x < \alpha\}$.

Theorem 1.24 (10.1)

Every nonempty finite ordered set has the order type of a section $\{1, \dots, n\}$; hence it is well-ordered.

Proof. Induction on the cardinality: remove the largest element (which exists in a finite ordered set, by induction) and append it at $n+1$. ■

Theorem 1.25 (Well-ordering theorem, Zermelo)

Every set can be well-ordered. (Equivalent to the axiom of choice.)

We accept this; a proof is a transfinite version of Theorem 1.22 using a choice function. In particular there exist uncountable well-ordered sets.

Lemma 1.26 (10.2)

There is a well-ordered set A with a largest element Ω such that the section S_Ω is uncountable but every other section of A is countable.

Proof. Let B be an uncountable well-ordered set and $C = \{1, 2\} \times B$ in dictionary order; some section of C is uncountable (e.g. that of $2 \times b$). Let Ω be the least element of C whose section is uncountable, and $A = S_\Omega \cup \{\Omega\}$. ■

S_Ω is called the **minimal uncountable well-ordered set**, and $\overline{S}_\Omega = S_\Omega \cup \{\Omega\}$.

Theorem 1.27 (10.3)

Every countable subset A of S_Ω has an upper bound in S_Ω.

Proof. $\bigcup_{a \in A} S_a$ is a countable union of countable sets, hence countable, so it is not all of S_Ω; any x outside it is an upper bound for A. ■

1.11 The maximum principle

Motivation. Zorn's lemma, equivalent to the axiom of choice, proves existence of maximal objects and is used for the Tychonoff theorem.

A **strict partial order** $\prec$ is nonreflexive and transitive; a subset on which it is a simple order is a **chain**.

Theorem 1.28 (11.1, Hausdorff maximum principle)

In a set with a strict partial order there is a maximal chain (a chain not properly contained in any other chain).

Proof. Well-order the set A by $<$ and define by transfinite recursion $h(\alpha) = \mathit{true}$ iff $\{\beta < \alpha : h(\beta) = \mathit{true}\} \cup \{\alpha\}$ is a chain for $\prec$. The set B of α with $h(\alpha)$ true is a chain (any two of its elements lie in the chain tested at the later one). It is maximal: if $B \cup \{\alpha\}$ were a chain for some $\alpha \notin B$, then $\{\beta < \alpha : \beta \in B\} \cup \{\alpha\}$ would be a chain, forcing $\alpha \in B$. ■

Theorem 1.29 (Zorn's lemma)

If A is strictly partially ordered and every chain has an upper bound in A, then A has a maximal element.

Proof. Take a maximal chain B and an upper bound c of B. If $c \prec d$ for some d, then $B \cup \{d\}$ is a chain properly containing B (note $d \notin B$ since $d \succ c \succeq$ every element of B), contradicting maximality. ■

Example 1.1

Every vector space has a basis: linearly independent sets under proper inclusion satisfy Zorn's hypothesis (a union of a chain is independent), and a maximal independent set spans. Zorn's lemma, the well-ordering theorem, the Hausdorff maximum principle and the axiom of choice are equivalent.

Chapter 2. Topological Spaces and Continuous Functions

2.1 Topological spaces

Motivation. A topology keeps exactly the properties of open sets in $\mathbb{R}^n$ that continuity and convergence use.

Definition (Topology)

A **topology** on a set X is a collection $\mathcal{T}$ of subsets of X, called **open sets**, such that $\emptyset, X \in \mathcal{T}$, arbitrary unions of members of $\mathcal{T}$ are in $\mathcal{T}$, and finite intersections of members of $\mathcal{T}$ are in $\mathcal{T}$.

Examples: the **discrete topology** $\mathcal{P}(X)$; the **trivial topology** $\{\emptyset, X\}$; the **finite complement topology** $\mathcal{T}_f = \{U : X - U \text{ finite}\} \cup \{\emptyset\}$ (a topology because $X - \bigcup U_\alpha = \bigcap (X - U_\alpha)$ and $X - \bigcap_{i \le n} U_i = \bigcup_{i \le n} (X - U_i)$ are finite); the countable complement topology. If $\mathcal{T} \subset \mathcal{T}'$ then $\mathcal{T}'$ is **finer** and $\mathcal{T}$ **coarser**.

2.2 Basis for a topology

Motivation. A basis, like the open discs in $\mathbb{R}^2$, generates all open sets, and most verifications need only be done on basis elements.

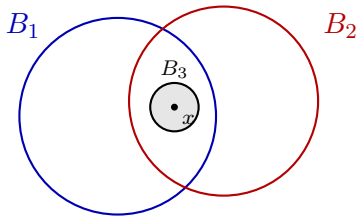


Figure. *The second basis axiom: if $x \in B_1 \cap B_2$, some basis element B_3 with $x \in B_3 \subset B_1 \cap B_2$ exists. This is what makes finite intersections of open sets open.*

Definition (Basis)

A collection $\mathcal{B}$ of subsets of X is a **basis** if (1) each x lies in at least one $B \in \mathcal{B}$, and (2) if $x \in B_1 \cap B_2$ then some $B_3 \in \mathcal{B}$ has $x \in B_3 \subset B_1 \cap B_2$. The topology **generated** by $\mathcal{B}$: U is open iff for each $x \in U$ there is $B \in \mathcal{B}$ with $x \in B \subset U$.

That this is a topology: unions are clear; for $U_1 \cap U_2$ and x in it, pick $B_i \ni x$ inside U_i and then $B_3 \subset B_1 \cap B_2$; finite intersections follow by induction. Examples: circular regions or rectangles in $\mathbb{R}^2$; one-point sets (discrete topology).

Lemma 2.1 (13.1)

If $\mathcal{B}$ is a basis for $\mathcal{T}$, then $\mathcal{T}$ equals the collection of all unions of elements of $\mathcal{B}$.

Proof. Unions of basis elements are open. An open U is the union of the basis elements B_x with $x \in B_x \subset U$, $x \in U$. ∎

Lemma 2.2 (13.2)

Let $\mathcal{C}$ be a collection of open sets of X such that for each open U and $x \in U$ there is $C \in \mathcal{C}$ with $x \in C \subset U$. Then $\mathcal{C}$ is a basis for the topology of X.

Proof. Condition (1) holds with $U = X$; for (2) apply the hypothesis to the open set $C_1 \cap C_2$. The topology $\mathcal{T}'$ generated by $\mathcal{C}$ contains $\mathcal{T}$ by hypothesis, and is contained in $\mathcal{T}$ because its sets are unions of members of $\mathcal{C} \subset \mathcal{T}$. ■

Lemma 2.3 (13.3)

Let $\mathcal{B}, \mathcal{B}'$ be bases for $\mathcal{T}, \mathcal{T}'$ on X. Then $\mathcal{T}'$ is finer than $\mathcal{T}$ iff for each x and each $B \in \mathcal{B}$ containing x there is $B' \in \mathcal{B}'$ with $x \in B' \subset B$.

Proof. ($\Leftarrow$) If $U \in \mathcal{T}$ and $x \in U$, choose B with $x \in B \subset U$, then B'; so $U \in \mathcal{T}'$.

($\Rightarrow$) $B \in \mathcal{T} \subset \mathcal{T}'$, and $\mathcal{T}'$ is generated by $\mathcal{B}'$. ■

On $\mathbb{R}$: the **standard topology** (basis: open intervals); the **lower limit topology** $\mathbb{R}_\ell$ (basis: $[a, b)$); the **K-topology** $\mathbb{R}_K$ (basis: (a, b) and $(a, b) - K$, $K = \{1/n : n \in \mathbb{Z}_+\}$).

Lemma 2.4 (13.4)

The topologies of $\mathbb{R}_\ell$ and $\mathbb{R}_K$ are strictly finer than the standard topology, but not comparable with one another.

Proof. Given $(a, b) \ni x$, both $[x, b)$ and (a, b) lie inside it, so both are finer (Lemma 2.3). $[x, d)$ contains no open interval around x, and $(-1, 1) - K$ contains no open interval around 0, so both are strictly finer. No basis element of $\mathbb{R}_K$ containing 10 lies in $[10, 11)$, and no basis element of $\mathbb{R}_\ell$ containing 0 lies in $(-1, 1) - K$. ■

A **subbasis** $\mathcal{S}$ is a collection of subsets whose union is X; the topology it generates is the collection of unions of finite intersections of members of $\mathcal{S}$ (these finite intersections form a basis).

2.3 The order topology

Motivation. Every linearly ordered set carries a topology generated by open intervals, giving test examples such as the ordered square.

Let X be simply ordered with more than one element. The **order topology** has as basis: all open intervals (a, b); all $[a_0, b)$ if X has a smallest element a_0; all $(a, b_0]$ if X has a largest element b_0. (If $x \in B_1 \cap B_2$, the intersection of two such sets is again of this form.) The open rays $(a, +\infty)$, $(-\infty, a)$ form a subbasis. Examples: $\mathbb{R}$ (standard topology); $\mathbb{Z}_+$ (discrete); $\mathbb{R} \times \mathbb{R}$ in the dictionary order (basis: vertical intervals $(a \times b, a \times d)$); $\{1, 2\} \times \mathbb{Z}_+$ in the dictionary order, which is not discrete since every neighbourhood of 2×1 contains points $1 \times n$.

2.4 The product topology on $X \times Y$

Motivation. Products of open sets form a basis for a topology on $X \times Y$.

The **product topology** on $X \times Y$ has basis $\mathcal{B} = \{U \times V : U \text{ open in } X,\ V \text{ open in } Y\}$ (a basis since $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$).

Theorem 2.5 (15.1)

If $\mathcal{B}$ and $\mathcal{C}$ are bases for X and Y, then $\{B \times C : B \in \mathcal{B}, C \in \mathcal{C}\}$ is a basis for the topology of $X \times Y$.

Proof. Given open W and $x \times y \in W$, choose $U \times V$ with $x \times y \in U \times V \subset W$, then $B \subset U$, $C \subset V$ basis elements containing x, y. Apply Lemma 2.2. ■

Theorem 2.6 (15.2)

$\mathcal{S} = \{\pi_1^{-1}(U) : U \text{ open in } X\} \cup \{\pi_2^{-1}(V) : V \text{ open in } Y\}$ is a subbasis for the product topology.

Proof. $\pi_1^{-1}(U) \cap \pi_2^{-1}(V) = U \times V$, so the finite intersections of members of $\mathcal{S}$ are exactly the basis sets $U \times V$ (together with sets of the same form). ■

2.5 The subspace topology

Motivation. A subset inherits a topology from its surroundings; "open" then depends on the ambient space.

For $Y \subset X$, the **subspace topology** is $\mathcal{T}_Y = \{Y \cap U : U \in \mathcal{T}\}$.

Lemma 2.7 (16.1)

If $\mathcal{B}$ is a basis for X, then $\{B \cap Y : B \in \mathcal{B}\}$ is a basis for the subspace topology on Y.

Proof. For U open in X and $y \in U \cap Y$, choose B with $y \in B \subset U$; then $y \in B \cap Y \subset U \cap Y$. Apply Lemma 2.2. ■

Lemma 2.8 (16.2)

If U is open in Y and Y is open in X, then U is open in X.

Proof. $U = Y \cap V$ with V open in X, an intersection of two open sets of X. ■

Theorem 2.9 (16.3)

If A is a subspace of X and B of Y, the product topology on $A \times B$ equals the subspace topology from $X \times Y$.

Proof. $(U \times V) \cap (A \times B) = (U \cap A) \times (V \cap B)$; the left sides form a basis for the subspace topology (Lemma 2.7) and the right sides a basis for the product topology (Theorem 2.5). ■

Theorem 2.10 (16.4)

If X is ordered with the order topology and $Y \subset X$ is convex (for $a < b$ in Y, $(a, b) \subset Y$), then the order topology on Y equals the subspace topology.

Proof. The intersection of an open ray $(a, +\infty)$ of X with Y is Y, $\emptyset$, or an open ray of Y (convexity is used when $a \notin Y$: then a is a lower or upper bound of Y). So subbasis elements of the subspace topology are open in the order topology. Conversely, an open ray of Y is the intersection of Y with the corresponding ray of X. ■

Without convexity the two can differ: in $Y = [0,1) \cup \{2\} \subset \mathbb{R}$, $\{2\}$ is open in the subspace topology but not in the order topology; in $I \times I$ with the dictionary order, the order topology differs from the subspace topology of $\mathbb{R} \times \mathbb{R}$ (dictionary order). We always give subsets of ordered sets the subspace topology unless said otherwise.

2.6 Closed sets and limit points

Motivation. Closures and limit points describe approximation; the Hausdorff condition restores uniqueness of limits.

A is **closed** if $X - A$ is open.

Theorem 2.11 (17.1)

$\emptyset$ and X are closed; arbitrary intersections and finite unions of closed sets are closed.

Proof. De Morgan's laws applied to the axioms for open sets. ■

Theorem 2.12 (17.2)

For a subspace Y, A is closed in Y iff $A = C \cap Y$ for some closed $C \subset X$.

Proof. If $A = C \cap Y$, then $Y - A = (X - C) \cap Y$ is open in Y. Conversely, $Y - A = U \cap Y$ with U open, and $A = (X - U) \cap Y$. ■

Theorem 2.13 (17.3)

If A is closed in Y and Y is closed in X, then A is closed in X.

The **interior** $\operatorname{Int} A$ is the union of the open sets contained in A; the **closure** $\overline{A}$ is the intersection of the closed sets containing A. So $\operatorname{Int} A \subset A \subset \overline{A}$, A is open iff $A = \operatorname{Int} A$, closed iff $A = \overline{A}$.

Theorem 2.14 (17.4)

If Y is a subspace of X, $A \subset Y$, and $\overline{A}$ is the closure of A in X, then the closure of A in Y is $\overline{A} \cap Y$.

Proof. $\overline{A} \cap Y$ is closed in Y and contains A, so it contains the closure B of A in Y. Conversely $B = C \cap Y$ with C closed in X (Theorem 2.12) and $C \supset A$, so $\overline{A} \subset C$ and $\overline{A} \cap Y \subset B$. ■

We say U is a **neighbourhood** of x if U is open and contains x, and that U **intersects** A if $U \cap A \neq \emptyset$.

Theorem 2.15 (17.5)

(a) $x \in \overline{A}$ iff every open set containing x intersects A.
(b) If the topology is given by a basis, $x \in \overline{A}$ iff every basis element containing x intersects A.

Proof. (a) Contrapositive: $x \notin \overline{A}$ iff some open $U \ni x$ misses A (take $U = X - \overline{A}$; conversely $X - U$ is a closed set containing A and missing x).

(b) Every open set containing x contains a basis element containing x. ■

For example, in $\mathbb{R}$ the closure of $(0,1]$ is $[0,1]$, of $\mathbb{Q}$ is $\mathbb{R}$, of $\{1/n\}$ is $\{0\} \cup \{1/n\}$. The point x is a **limit point** of A if every neighbourhood of x intersects A in a point other than x; equivalently $x \in \overline{A - \{x\}}$. Let A' denote the set of limit points.

Theorem 2.16 (17.6)

$\overline{A} = A \cup A'$.

Proof. $A' \subset \overline{A}$ by Theorem 2.15. If $x \in \overline{A} - A$, every neighbourhood of x meets A in a point, which is not x; so $x \in A'$. ■

Corollary 2.17 (17.7)

A subset is closed iff it contains all its limit points.

A sequence (x_n) **converges** to x if for every neighbourhood U of x there is N with $x_n \in U$ for $n \geq N$. In general limits need not be unique. X is a **Hausdorff space** if distinct points have disjoint neighbourhoods.

Theorem 2.18 (17.8)

Every finite point set in a Hausdorff space is closed.

Proof. For a single point x_0: any $x \neq x_0$ has a neighbourhood missing a neighbourhood of x_0, in particular missing x_0, so $x \notin \overline{\{x_0\}}$. Finite unions of closed sets are closed. ■

The condition that finite sets are closed is the T_1 **axiom**; it is weaker than Hausdorff (finite complement topology on $\mathbb{R}$).

Theorem 2.19 (17.9)

In a T_1 space, x is a limit point of A iff every neighbourhood of x contains infinitely many points of A.

Proof. ($\Rightarrow$) If a neighbourhood U of x met A in finitely many points, and $x_1, \ldots, x_m$ are those different from x, then $U - \{x_1, \ldots, x_m\}$ is a neighbourhood of x (finite sets are closed) meeting $A - \{x\}$ not at all.

($\Leftarrow$) is clear. ■

Theorem 2.20 (17.10)

In a Hausdorff space a sequence converges to at most one point.

Proof. If $x_n \to x$ and $y \neq x$, take disjoint neighbourhoods $U \ni x$, $V \ni y$; U contains all but finitely many x_n, so V cannot, and $x_n \not\to y$. ■

Theorem 2.21 (17.11)

Every simply ordered set is Hausdorff in the order topology; products of two Hausdorff spaces and subspaces of Hausdorff spaces are Hausdorff.

Proof. For $a < b$ in an ordered set: if some c lies between them, $(-\infty, c)$ and $(c, +\infty)$ separate; otherwise $(-\infty, b)$ and $(a, +\infty)$ do. For products use $U_1 \times V_1$ and $U_2 \times V_2$ (separating in one coordinate suffices); for subspaces intersect with the subspace. ■

2.7 Continuous functions

Motivation. Continuity means preimages of open sets are open; below are the standard ways to build continuous maps.

$f : X \to Y$ is **continuous** if $f^{-1}(V)$ is open for every open $V \subset Y$; it suffices to check V in a basis or subbasis, since preimages preserve unions and intersections.

Theorem 2.22 (18.1)

For $f : X \to Y$ the following are equivalent:
(1) f is continuous;
(2) $f(\overline{A}) \subset \overline{f(A)}$ for every $A \subset X$;
(3) $f^{-1}(B)$ is closed for every closed $B \subset Y$;
(4) for each x and each neighbourhood V of $f(x)$ there is a neighbourhood U of x with $f(U) \subset V$.

Proof. (1)⇒(2): for $x \in \overline{A}$ and a neighbourhood V of $f(x)$, $f^{-1}(V)$ is a neighbourhood of x meeting A in some y, and $f(y) \in V \cap f(A)$. (2)⇒(3): with $A = f^{-1}(B)$, $f(\overline{A}) \subset \overline{f(A)} \subset \overline{B} = B$, so $\overline{A} \subset A$. (3)⇒(1): complements. (1)⇒(4): $U = f^{-1}(V)$. (4)⇒(1): $f^{-1}(V)$ is the union of the neighbourhoods U_x given by (4). ■

A **homeomorphism** is a continuous bijection with continuous inverse; equivalently a bijection with $f(U)$ open iff U open. A continuous injection $f : X \to Y$ that is a homeomorphism onto $f(X)$ (subspace topology) is an **imbedding**. Examples: the identity $\mathbb{R}_\ell \to \mathbb{R}$ is a continuous bijection but not a homeomorphism; $[0, 1) \to S^1$, $t \mapsto (\cos 2\pi t, \sin 2\pi t)$ is a continuous bijection that is not a homeomorphism.

Theorem 2.23 (18.2, rules for constructing continuous functions)

(a) Constant functions are continuous.
(b) Inclusions of subspaces are continuous.
(c) Composites of continuous functions are continuous.
(d) Restricting the domain of a continuous function to a subspace keeps it continuous.
(e) Restricting or expanding the range: if $f : X \to Y$ is continuous and $Z \subset Y$ contains $f(X)$, then $f : X \to Z$ is continuous; if $Y \subset Z$ is a subspace, $f : X \to Z$ is continuous.
(f) Local formulation: f is continuous if X is a union of open sets U_α with each $f|_{U_\alpha}$ continuous.

Proof. (a) Preimages are $\emptyset$ or X.

(b) $j^{-1}(U) = U \cap A$.

(c) $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$.

(d) Composite with the inclusion.

(e) $f^{-1}(Z \cap V) = f^{-1}(V)$; and composite with the inclusion $Y \to Z$.

(f) $f^{-1}(V) = \bigcup (f|_{U_\alpha})^{-1}(V)$, a union of sets open in U_α, hence open in X (Lemma 2.8). ■

Theorem 2.24 (18.3, the pasting lemma)

Let $X = A \cup B$ with A, B closed, $f : A \to Y$ and $g : B \to Y$ continuous and $f = g$ on $A \cap B$. Then $h = f \cup g : X \to Y$ is continuous. (The same holds with A, B open.)

Proof. For closed $C \subset Y$, $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$; each is closed in A or B, hence in X (Theorem 2.13). ∎

Theorem 2.25 (18.4, maps into products)

$f : A \to X \times Y$, $f = (f_1, f_2)$, is continuous iff f_1 and f_2 are continuous.

Proof. The projections are continuous ($\pi_1^{-1}(U) = U \times Y$), so $f_i = \pi_i \circ f$ are continuous if f is. Conversely $f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V)$. ∎

2.8 The product topology

Motivation. For infinite products the product topology, not the box topology, makes a map continuous exactly when each coordinate is.

For an indexed family $\{X_\alpha\}_{\alpha \in J}$, the **box topology** on $\prod X_\alpha$ has basis all products $\prod U_\alpha$ of open sets, and the **product topology** has subbasis $\mathcal{S} = \bigcup_\beta \{\pi_\beta^{-1}(U_\beta)\}$.

Theorem 2.26 (19.1, comparison)

The box topology has basis all $\prod U_\alpha$ with U_α open; the product topology has basis all $\prod U_\alpha$ with U_α open and $U_\alpha = X_\alpha$ for all but finitely many α. For finite products they coincide; in general the box topology is finer.

Proof. A finite intersection $\pi_{\beta_1}^{-1}(U_{\beta_1}) \cap \cdots \cap \pi_{\beta_n}^{-1}(U_{\beta_n})$ is exactly a product $\prod U_\alpha$ with $U_\alpha = X_\alpha$ except for $\alpha \in \{\beta_1, \ldots, \beta_n\}$. ∎

Theorem 2.27 (19.2)

If each X_α has a basis $\mathcal{B}_\alpha$, products $\prod B_\alpha$ with $B_\alpha \in \mathcal{B}_\alpha$ form a basis for the box topology, and those with $B_\alpha \in \mathcal{B}_\alpha$ for finitely many α and $B_\alpha = X_\alpha$ otherwise form a basis for the product topology.

Theorem 2.28 (19.3)

If $A_\alpha \subset X_\alpha$, then $\prod A_\alpha$ is a subspace of $\prod X_\alpha$ when both carry the box topology, or both the product topology.

Theorem 2.29 (19.4)

A product of Hausdorff spaces is Hausdorff in both topologies.

Theorem 2.30 (19.5)

In either topology, $\prod \overline{A_\alpha} = \overline{\prod A_\alpha}$.

Proof. If $x \in \prod \overline{A_\alpha}$ and $U = \prod U_\alpha$ is a basis element containing x, choose $y_\alpha \in U_\alpha \cap A_\alpha$; then $y \in U \cap \prod A_\alpha$. Conversely, if $x \in \overline{\prod A_\alpha}$ and $V_\beta \ni x_\beta$ is open, $\pi_\beta^{-1}(V_\beta)$ meets $\prod A_\alpha$, so V_β meets A_β. (Theorems 2.27–2.29 are proved like their two-factor versions.) ∎

Theorem 2.31 (19.6)

Let $f : A \to \prod X_\alpha$, $f(a) = (f_\alpha(a))$, where $\prod X_\alpha$ has the product topology. Then f is continuous iff each f_α is continuous.

Proof. π_β is continuous, giving necessity. Conversely, $f^{-1}(\pi_\beta^{-1}(U_\beta)) = f_\beta^{-1}(U_\beta)$ is open, and these sets form a subbasis. ■

This fails for the box topology: $f : \mathbb{R} \to \mathbb{R}^\omega$, $f(t) = (t, t, \dots)$, has continuous coordinates, but $f^{-1}(\prod(-\frac{1}{n}, \frac{1}{n})) = \{0\}$. Henceforth products carry the product topology.

2.9 The metric topology

Motivation. The standard metrics on $\mathbb{R}^n$ give the product topology, and the uniform metric on $\mathbb{R}^J$ lies strictly between the product and box topologies.

A **metric** d on X satisfies $d(x, y) \geq 0$ with equality iff $x = y$, symmetry, and the triangle inequality. The ϵ-balls $B_d(x, \epsilon)$ form a basis (if $y \in B_d(x, \epsilon)$, then $B_d(y, \epsilon - d(x, y)) \subset B_d(x, \epsilon)$), generating the **metric topology**; U is open iff each $y \in U$ has a ball $B_d(y, \delta) \subset U$. X is **metrizable** if some metric induces its topology. A is **bounded** if $\sup_{a,b\in A} d(a, b) = \operatorname{diam} A < \infty$.

Theorem 2.32 (20.1)

$\bar{d}(x, y) = \min\{d(x, y), 1\}$ is a metric inducing the same topology as d (the **standard bounded metric**).

Proof. Triangle inequality: if $d(x, y) \geq 1$ or $d(y, z) \geq 1$ the right side is at least 1; otherwise $\bar{d}(x, z) \leq d(x, z) \leq d(x, y) + d(y, z) = \bar{d}(x, y) + \bar{d}(y, z)$. The balls of radius < 1 coincide and form a basis for both topologies. ■

Lemma 2.33 (20.2)

For metrics d, d' on X with topologies $\mathcal{T}, \mathcal{T}'$, $\mathcal{T}'$ is finer than $\mathcal{T}$ iff for each x and $\epsilon > 0$ there is $\delta > 0$ with $B_{d'}(x, \delta) \subset B_d(x, \epsilon)$.

Proof. This is Lemma 2.3 for the bases of balls, using that every ball containing x contains a ball centred at x. ■

Theorem 2.34 (20.3)

The euclidean metric d and the square metric $\rho(x, y) = \max|x_i - y_i|$ on $\mathbb{R}^n$ both induce the product topology.

Proof. $\rho \leq d \leq \sqrt{n}\rho$ gives $B_d(x, \epsilon) \subset B_\rho(x, \epsilon)$ and $B_\rho(x, \epsilon/\sqrt{n}) \subset B_d(x, \epsilon)$, so they agree. Product basis elements $\prod(a_i, b_i) \ni x$ contain $B_\rho(x, \epsilon)$ for $\epsilon = \min_i \min(x_i - a_i, b_i - x_i)$, and $B_\rho(x, \epsilon)$ is itself a product of intervals. ■

On $\mathbb{R}^J$ the **uniform metric** is $\bar{\rho}(x, y) = \sup_\alpha \bar{d}(x_\alpha, y_\alpha)$.

Theorem 2.35 (20.4)

The uniform topology on $\mathbb{R}^J$ is finer than the product topology and coarser than the box topology; if J is infinite, all three are different.

Proof. A product basis element $\prod U_\alpha$ restricting $\alpha_1, \ldots, \alpha_n$ contains a ball $B_{\bar{\rho}}(x, \epsilon)$ (ϵ the minimum of the radii needed in those coordinates, and < 1). A ball $B_{\bar{\rho}}(x, \epsilon)$ contains the box set $\prod(x_\alpha - \epsilon/2, x_\alpha + \epsilon/2)$. For infinite J: $B_{\bar{\rho}}(0, 1)$ is not product-open, and $\prod(-1/n, 1/n)$ (indexing a countable subset) contains no uniform ball around 0. ■

Theorem 2.36 (20.5)

With $\bar{d}(a, b) = \min\{|a - b|, 1\}$, $D(x, y) = \sup_i \bar{d}(x_i, y_i)/i$ is a metric inducing the product topology on $\mathbb{R}^\omega$.

Proof. D is a metric (sup of metrics divided by i). Given U open in the product topology and $x \in U$, choose a basis set $\prod_{i \le N}(x_i - \epsilon, x_i + \epsilon) \times \mathbb{R} \times \cdots \subset U$, $\epsilon < 1$; if $D(x, y) < \epsilon/N$ then $\bar{d}(x_i, y_i) < \epsilon$ for $i \le N$, so $B_D(x, \epsilon/N) \subset U$. Conversely, given $B_D(x, \epsilon)$, choose N with $1/N < \epsilon$ and let $V = \prod_{i<N}(x_i - \epsilon/2, x_i + \epsilon/2) \times \mathbb{R} \times \mathbb{R} \times \cdots$. For $y \in V$, $\bar{d}(x_i, y_i)/i \le \epsilon/2$ for $i < N$ and $\le 1/N$ for $i \ge N$, so $D(x, y) \le \max(\epsilon/2, 1/N) < \epsilon$ and $V \subset B_D(x, \epsilon)$. ■

2.10 The metric topology (continued)

Motivation. In metric spaces sequences detect closure and continuity, and uniform limits of continuous functions are continuous; both can fail in general spaces.

Subspaces of metric spaces are metric spaces (restrict d); order topologies may or may not be metrizable; Hausdorff is necessary for metrizability.

Theorem 2.37 (21.1)

For metrizable X, Y with metrics d_X, d_Y, $f : X \to Y$ is continuous iff for each x and $\epsilon > 0$ there is $\delta > 0$ with $d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \epsilon$.

Proof. Apply Theorem 2.22(4) with neighbourhoods taken to be balls. ■

Lemma 2.38 (21.2, the sequence lemma)

If a sequence of points of A converges to x, then $x \in \overline{A}$; the converse holds if X is metrizable (indeed if X has a countable neighbourhood basis at x).

Proof. Every neighbourhood of x contains points of the sequence. Conversely, if $x \in \overline{A}$ and X is metrizable, choose $x_n \in B(x, 1/n) \cap A$; then $x_n \to x$. ■

Theorem 2.39 (21.3)

If f is continuous, then $x_n \to x$ implies $f(x_n) \to f(x)$; the converse holds if X is metrizable.

Proof. If $V \ni f(x)$ is open, $f^{-1}(V) \ni x$ contains x_n for large n. Conversely, for metrizable X we show $f(\overline{A}) \subset \overline{f(A)}$: if $x \in \overline{A}$, choose $x_n \in A$ with $x_n \to x$ (Lemma 2.38); then $f(x_n) \to f(x)$, so $f(x) \in \overline{f(A)}$. Apply Theorem 2.22. ■

In particular $\mathbb{R}^\omega$ with the box topology and $\mathbb{R}^J$ (J uncountable) in the product topology are not metrizable: in both, a point lies in the closure of a set without being a limit of a sequence from it.

Lemma 2.40 (21.4)

Addition, subtraction and multiplication $\mathbb{R}\times\mathbb{R}\to\mathbb{R}$ and division $\mathbb{R}\times(\mathbb{R}-\{0\})\to\mathbb{R}$ are continuous.

Proof. Standard ϵ–δ estimates using the metric $|a-a'|+|b-b'|$ on $\mathbb{R}^2$; e.g. $|ab-a'b'|\le|a||b-b'|+|b'||a-a'|$. ■

Theorem 2.41 (21.5)

If $f,g:X\to\mathbb{R}$ are continuous, so are $f\pm g$, fg, and f/g where $g\neq 0$.

Proof. $f+g$ is the composite of $(f,g):X\to\mathbb{R}\times\mathbb{R}$ (Theorem 2.25) and addition (Lemma 2.40). ■

$f_n:X\to Y$ (Y metric) **converges uniformly** to f if for every ϵ there is N with $d(f_n(x),f(x))<\epsilon$ for all $n\ge N$ and all x.

Theorem 2.42 (21.6, uniform limit theorem)

If continuous $f_n:X\to Y$ converge uniformly to f, then f is continuous.

Proof. Let V be open in Y, $x_0\in f^{-1}(V)$, and $B(f(x_0),\epsilon)\subset V$. Choose N with $d(f_N(x),f(x))<\epsilon/3$ for all x, and a neighbourhood U of x_0 with $f_N(U)\subset B(f_N(x_0),\epsilon/3)$. For $x\in U$, $d(f(x),f(x_0))<\epsilon$, so $f(U)\subset V$. ■

2.11 The quotient topology

Motivation. The quotient topology formalizes gluing, and its universal property decides continuity of maps on the glued space.

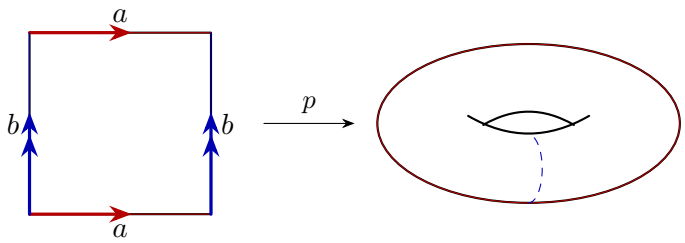


Figure. *Gluing opposite sides of a square (identify the edges labelled a, and those labelled b, respecting arrows) gives the torus; the quotient map p carries the quotient topology.*

A surjective map $p:X\to Y$ is a **quotient map** if $U\subset Y$ is open iff $p^{-1}(U)$ is open. Equivalently, p is continuous and maps **saturated** open sets (sets of the form $p^{-1}(V)$) to open sets. Open or closed continuous surjections are quotient maps (e.g. the projection $\pi_1:X\times Y\to X$ is open). If X is a space and $p:X\to A$ a surjection onto a set, the **quotient topology** on A is the unique topology making p a quotient map; for a partition X^* of X we get the **quotient space** X^*.

Theorem 2.43 (22.1)

Let $p : X \to Y$ be a quotient map, $A \subset X$ saturated, and $q = p|_A : A \to p(A)$.
(1) If A is open or closed, q is a quotient map.
(2) If p is open or closed, q is a quotient map.

Proof. For $V \subset p(A)$, saturation gives $q^{-1}(V) = p^{-1}(V)$ and $p(U \cap A) = p(U) \cap p(A)$ for U open (or closed).

(1) If A is open and $q^{-1}(V)$ is open in A, it is open in X, so $p^{-1}(V)$ is open and V is open in Y, hence in $p(A)$.

(2) If p is open and $q^{-1}(V) = U \cap A$ with U open, then $V = q(U \cap A) = p(U) \cap p(A)$ is open in $p(A)$. Closed cases are the same with closed sets. ■

Composites of quotient maps are quotient maps; products of quotient maps need not be; quotients of Hausdorff spaces need not be Hausdorff.

Theorem 2.44 (22.2)

Let $p : X \to Y$ be a quotient map and $g : X \to Z$ constant on each fibre $p^{-1}(y)$. Then g induces $f : Y \to Z$ with $f \circ p = g$; f is continuous iff g is, and f is a quotient map iff g is.

Proof. f is well defined. For $V \subset Z$ open, $g^{-1}(V) = p^{-1}(f^{-1}(V))$ is open iff $f^{-1}(V)$ is open (quotient map). If g is a quotient map, f is surjective, and $f^{-1}(V)$ open implies $g^{-1}(V)$ open, hence V open; conversely composites of quotient maps are quotient maps. ■

Corollary 2.45 (22.3)

Let $g : X \to Z$ be a continuous surjection and $X^* = \{g^{-1}(z)\}$ with the quotient topology.
(a) g induces a continuous bijection $f : X^* \to Z$, which is a homeomorphism iff g is a quotient map.
(b) If Z is Hausdorff, so is X^*.

Proof. (a) is Theorem 2.44.

(b) Disjoint open sets in Z pull back by the continuous f to disjoint open sets in X^*. ■

Example 2.1

(a) Collapsing ∂B^2 to a point gives S^2; identifying opposite edges of the square gives the torus.
(b) $p : \mathbb{R} \to S^1$, $p(t) = (\cos 2\pi t, \sin 2\pi t)$ is a quotient map (it is open), so $\mathbb{R}/\mathbb{Z} \cong S^1$ by Corollary 2.45.
(c) For $\mathbb{Q} \subset \mathbb{R}$ collapsed to a point, $p \times \mathrm{id}_{\mathbb{Q}} : \mathbb{R} \times \mathbb{Q} \to \mathbb{R}^* \times \mathbb{Q}$ is not a quotient map (Munkres Example 22.7).

Chapter 3. Connectedness and Compactness

3.1 Connected spaces

Motivation. Connectedness abstracts the fact that $[a, b]$ does not split into two disjoint open pieces, and gives the intermediate value theorem in any setting.

A **separation** of X is a pair U, V of disjoint nonempty open sets with $X = U \cup V$. X is **connected** if no separation exists; equivalently, the only sets both open and closed are $\emptyset$ and X.

Lemma 3.1 (23.1)

If $Y \subset X$, a separation of Y is a pair of disjoint nonempty sets A, B with $A \cup B = Y$, neither containing a limit point of the other.

Proof. If A, B separate Y, A is closed in Y, so $A = \overline{A} \cap Y$ (Theorem 2.14) and $\overline{A} \cap B = \emptyset$: B contains no limit point of A; symmetrically. Conversely, if $\overline{A} \cap B = \emptyset = A \cap \overline{B}$, then $\overline{A} \cap Y = A$ and $\overline{B} \cap Y = B$ are closed in Y, so their complements B, A are open in Y. ■

Examples: the indiscrete topology is connected; $\mathbb{Q}$ is totally disconnected (its only connected subspaces are points); $Y = [-1, 0) \cup (0, 1]$ is disconnected.

Lemma 3.2 (23.2)

If C, D separate X and $Y \subset X$ is connected, then $Y \subset C$ or $Y \subset D$.

Proof. $C \cap Y$ and $D \cap Y$ are open in Y and disjoint with union Y; if both were nonempty they would separate Y. ■

Theorem 3.3 (23.3)

The union of a collection of connected subspaces of X with a point in common is connected.

Proof. Let p be the common point and suppose C, D separate $Y = \bigcup A_\alpha$, with $p \in C$. Each A_α lies in C or D (Lemma 3.2), hence in C as $p \in A_\alpha$. So $D = \emptyset$, a contradiction. ■

Theorem 3.4 (23.4)

If A is connected and $A \subset B \subset \overline{A}$, then B is connected.

Proof. If C, D separate B, then $A \subset C$, say (Lemma 3.2); then $\overline{A} \subset \overline{C}$, and $\overline{C} \cap D = \emptyset$ (Lemma 3.1), so $B \cap D \subset \overline{A} \cap D = \emptyset$. ■

Theorem 3.5 (23.5)

The image of a connected space under a continuous map is connected.

Proof. If A, B separate $f(X)$, then $f^{-1}(A), f^{-1}(B)$ separate X. ■

Theorem 3.6 (23.6)

A finite cartesian product of connected spaces is connected.

Proof. For $X \times Y$ fix $a \times b$. For each x, $T_x = (X \times b) \cup (x \times Y)$ is connected (Theorem 3.3, with common point $x \times b$). All T_x contain $a \times b$, and their union is $X \times Y$. Induct for more factors. ■

Arbitrary products are connected in the product topology (the set of points differing from a fixed point in finitely many coordinates is a connected dense subset), but $\mathbb{R}^\omega$ in the box topology is not: bounded and unbounded sequences form a separation.

3.2 Connected subspaces of the real line

Motivation. The least upper bound property makes $\mathbb{R}$ and its intervals connected; path connectedness then covers $\mathbb{R}^n$ and spheres.

A simply ordered set L with more than one element is a **linear continuum** if it has the least upper bound property and for $x < y$ there is z with $x < z < y$.

Theorem 3.7 (24.1)

A linear continuum L in the order topology is connected, and so are its intervals and rays.

Proof. It suffices to show convex subsets Y are connected. Suppose A, B separate Y, $a \in A$, $b \in B$, $a < b$. $[a, b] \subset Y$, so $A_0 = A \cap [a, b]$, $B_0 = B \cap [a, b]$ separate $[a, b]$. Let $c = \sup A_0$. If $c \in B_0$, then $c \neq a$, so $c > a$; B_0 is open in $[a, b]$, so some $(d, c] \subset B_0$, and then d is a smaller upper bound for A_0 (since $(d, c]$ misses A_0 and c is the sup): contradiction. If $c \in A_0$, then $c < b$ and some $[c, e) \subset A_0$; choosing z with $c < z < e$ gives $z \in A_0$, contradicting $c = \sup A_0$. ■

Corollary 3.8 (24.2)

$\mathbb{R}$ is connected, and so are its intervals and rays.

Theorem 3.9 (24.3, intermediate value theorem)

Let $f : X \to Y$ be continuous, X connected, Y ordered in the order topology. If $a, b \in X$ and r lies between $f(a)$ and $f(b)$, then $f(c) = r$ for some $c \in X$.

Proof. Otherwise $f(X) \cap (-\infty, r)$ and $f(X) \cap (r, +\infty)$ separate the connected set $f(X)$. ■

Other linear continua: $I \times I$ in the dictionary order (the ordered square), and the **long line** $L = S_\Omega \times [0, 1)$. A **path** from x to y is a continuous $f : [a, b] \to X$ with $f(a) = x$, $f(b) = y$; X is **path connected** if every two points are joined by a path. Path connected spaces are connected (Theorems 3.3, 3.5, 3.8).

Example 3.1

(a) The unit ball B^n and punctured space $\mathbb{R}^n - \{0\}$ $(n > 1)$ are path connected, hence so is S^{n-1} (its image under $x \mapsto x/\|x\|$).
(b) The ordered square is connected but not path connected: a path from the least to the greatest element would be surjective, and the uncountably many disjoint open intervals $U_x = (x \times 0, x \times 1)$ would have disjoint open preimages each containing a rational.
(c) The **topologist's sine curve** $\overline{S}$, the closure in $\mathbb{R}^2$ of $S = \{(x, \sin(1/x)) : 0 < x \leq 1\}$, is connected (Theorem 3.4) but not path connected: if $f : [a, c] \to \overline{S}$ were a path from 0×0 into S, let $b = \sup\{t : f(t) \in 0 \times [-1, 1]\}$; on $(b, c]$, $f(t) \in S$, and choosing $t_n \downarrow b$ with $\sin(1/x(t_n)) = (-1)^n$ contradicts the continuity of the second coordinate at b.

3.3 Components and local connectedness

Motivation. Components are the maximal connected pieces; counting them distinguishes, for example, $\mathbb{R}$ from $\mathbb{R}^2$.

Define $x \sim y$ if some connected subspace contains both; the classes are the **components** of X.

Theorem 3.10 (25.1)

The components of X are connected disjoint subspaces whose union is X, and each nonempty connected subspace meets only one of them.

Proof. $\sim$ is an equivalence relation (transitivity by Theorem 3.3). A connected A meeting components C_1, C_2 in x_1, x_2 gives $x_1 \sim x_2$. For connectedness of a component C, fix $x_0 \in C$; each $x \in C$ lies in a connected $A_x \ni x_0$, contained in C by the previous sentence, and $C = \bigcup A_x$ is connected by Theorem 3.3. ■

Similarly, define $x \sim y$ if a path joins them; the classes are the **path components**.

Theorem 3.11 (25.2)

The path components of X are path-connected disjoint subspaces with union X, and each nonempty path-connected subspace meets only one of them.

Proof. Transitivity: concatenate paths $f : [a, b] \to X$ and $g : [b, c] \to X$ via the pasting lemma. ■

Components are closed (the closure of a connected set is connected); path components need not be (the topologist's sine curve has two path components, S and $0 \times [-1, 1]$, the first not closed). X is **locally connected at** x if every neighbourhood of x contains a connected neighbourhood of x; similarly **locally path connected**. Intervals in $\mathbb{R}$ are both; $\mathbb{Q}$ is neither; the topologist's sine curve is connected but not locally connected.

Theorem 3.12 (25.3)

X is locally connected iff for every open U each component of U is open in X.

Proof. ($\Rightarrow$) If C is a component of U and $x \in C$, a connected neighbourhood $V \subset U$ of x lies in C.

($\Leftarrow$) For a neighbourhood U of x, the component of U containing x is a connected neighbourhood inside U. ■

Theorem 3.13 (25.4)

X is locally path connected iff for every open U each path component of U is open in X.

Theorem 3.14 (25.5)

Each path component of X lies in a component. If X is locally path connected, components and path components coincide.

Proof. The first claim since path connected sets are connected. Let C be a component and $P \subset C$ a path component. The path components of X are open (Theorem 3.13 with $U = X$); Q, the union of the other path components meeting C, is open, so P, Q would separate C unless $Q = \emptyset$. Hence $P = C$. ■

3.4 Compact spaces

Motivation. Compactness, every open cover has a finite subcover, passes local information to global and is preserved by continuous maps and products.

A collection of subsets **covers** X if its union is X; it is an **open covering** if its members are open. X is **compact** if every open covering contains a finite subcollection covering X. Examples: $\mathbb{R}$ is not compact; $\{0\} \cup \{1/n\}$ is compact; finite spaces and spaces with the finite complement topology are compact.

Lemma 3.15 (26.1)

A subspace $Y \subset X$ is compact iff every covering of Y by sets open in X contains a finite subcollection covering Y.

Proof. Open sets of Y are intersections $Y \cap A$ with A open in X; translate coverings in both directions. ■

Theorem 3.16 (26.2)

Every closed subspace of a compact space is compact.

Proof. Given a covering $\mathcal{A}$ of a closed Y by sets open in X, $\mathcal{A} \cup \{X - Y\}$ covers X; a finite subcover, minus $X - Y$, covers Y. ■

Theorem 3.17 (26.3)

Every compact subspace of a Hausdorff space is closed.

Lemma 3.18 (26.4)

If Y is a compact subspace of a Hausdorff space X and $x_0 \notin Y$, there are disjoint open $U \ni x_0$ and $V \supset Y$.

Proof. For each $y \in Y$ choose disjoint neighbourhoods $U_y \ni x_0$, $V_y \ni y$. Finitely many $V_{y_1}, \ldots, V_{y_n}$ cover Y; take $V = \bigcup V_{y_i}$ and $U = \bigcap U_{y_i}$. This proves Theorem 3.17 too: $X - Y$ is open. ■

Theorem 3.19 (26.5)

The image of a compact space under a continuous map is compact.

Proof. If $\mathcal{A}$ covers $f(X)$ by open sets of Y, $\{f^{-1}(A)\}$ covers X; finitely many suffice. ■

Theorem 3.20 (26.6)

A continuous bijection $f : X \to Y$ from a compact space to a Hausdorff space is a homeomorphism.

Proof. For closed $A \subset X$: A is compact (Theorem 3.16), $f(A)$ is compact (3.19), hence closed (3.17). So f^{-1} is continuous. ■

Lemma 3.21 (26.8, the tube lemma)

If Y is compact and N is an open set of $X \times Y$ containing the slice $x_0 \times Y$, then N contains a tube $W \times Y$ with W a neighbourhood of x_0.

Proof. Cover $x_0 \times Y$ by basis elements $U \times V \subset N$ meeting it; it is compact (homeomorphic to Y), so finitely many $U_i \times V_i$ cover it. Let $W = \bigcap U_i$. Every $x \times y$ with $x \in W$ lies in some $U_i \times V_i$ (choose i with $y \in V_i$), so $W \times Y \subset N$. ■

Theorem 3.22 (26.7)

A product of finitely many compact spaces is compact.

Proof. Let $\mathcal{A}$ be an open cover of $X \times Y$. For each x_0, finitely many members cover $x_0 \times Y$; their union N contains a tube $W_{x_0} \times Y$. Finitely many W_{x_i} cover X, and the corresponding finitely many members of $\mathcal{A}$ cover $X \times Y$. ■

A collection $\mathcal{C}$ has the **finite intersection property** if every finite subcollection has nonempty intersection.

Theorem 3.23 (26.9)

X is compact iff every collection $\mathcal{C}$ of closed sets with the finite intersection property has $\bigcap_{C \in \mathcal{C}} C \neq \emptyset$.

Proof. Pass to complements: $\{X - C\}$ is an open cover iff $\bigcap C = \emptyset$, and a finite subcollection covers iff the corresponding finite intersection is empty. ■

In particular nested sequences $C_1 \supset C_2 \supset \cdots$ of nonempty closed sets in a compact space have nonempty intersection.

3.5 Compact subspaces of the real line

Motivation. $[a, b]$ is compact, hence so are closed bounded sets in $\mathbb{R}^n$; the extreme value theorem, Lebesgue number lemma and uniform continuity follow.

Theorem 3.24 (27.1)

If X is simply ordered with the least upper bound property, every closed interval $[a, b]$ in X is compact (order topology).

Proof. Let $\mathcal{A}$ be a covering of $[a, b]$ by sets open in $[a, b]$.

Step 1. for $x \in [a, b)$ there is $y > x$ such that $[x, y]$ is covered by at most two members of $\mathcal{A}$ (if x has an immediate successor y, use the members containing x, y; otherwise take $A \ni x$ containing $[x, c)$ and any $y \in (x, c)$).

Step 2. let C be the set of $y > a$ in $[a, b]$ such that $[a, y]$ is finitely covered; $C \neq \emptyset$ by Step 1. Let $c = \sup C$.

Step 3. $c \in C$: choose $A \ni c$ containing some $(d, c]$; if $c \notin C$, some $z \in C \cap (d, c)$, and $[a, z]$ plus A covers $[a, c]$.

Step 4. $c = b$: otherwise Step 1 extends past c. ■

Corollary 3.25 (27.2)

Every closed interval in $\mathbb{R}$ is compact.

Theorem 3.26 (27.3)

$A \subset \mathbb{R}^n$ is compact iff it is closed and bounded (in the euclidean or square metric).

Proof. If A is compact, it is closed (Theorem 3.17) and covered by finitely many open balls $B_\rho(0, m)$, hence bounded. Conversely, a closed bounded A lies in a cube $[-P, P]^n$, compact by Theorems 3.22 and 3.25; apply Theorem 3.16. ■

Theorem 3.27 (27.4, extreme value theorem)

If $f : X \to Y$ is continuous, X compact and Y ordered, there are c, d with $f(c) \leq f(x) \leq f(d)$ for all x.

Proof. $A = f(X)$ is compact. If A had no largest element, the rays $(-\infty, a)$, $a \in A$, would cover A; a finite subcover $(-\infty, a_i)$ misses the largest a_i. So A has a largest element $f(d)$; similarly a smallest. ■

For $A \subset X$ in a metric space, $d(x, A) = \inf_{a \in A} d(x, a)$ is continuous in x (indeed $|d(x, A) - d(y, A)| \leq d(x, y)$).

Lemma 3.28 (27.5, Lebesgue number lemma)

Let $\mathcal{A}$ be an open covering of a compact metric space X. There is $\delta > 0$ such that every subset of diameter less than δ lies in some member of $\mathcal{A}$.

Proof. If $X \in \mathcal{A}$ take any δ. Otherwise choose a finite subcover $A_1, \ldots, A_n$, let $C_i = X - A_i$ and $f(x) = \frac{1}{n} \sum d(x, C_i)$. $f > 0$ (each x lies in some A_i, so $d(x, C_i) > 0$) and continuous, so $f \geq \delta > 0$ (Theorem 3.27). If $\operatorname{diam} B < \delta$ and $x_0 \in B$, then $B \subset B(x_0, \delta)$; choosing m with $d(x_0, C_m)$ maximal gives $d(x_0, C_m) \geq f(x_0) \geq \delta$, so $B(x_0, \delta) \subset A_m$. ■

Theorem 3.29 (27.6, uniform continuity theorem)

A continuous map from a compact metric space to a metric space is uniformly continuous.

Proof. Given ϵ, cover X by the open sets $f^{-1}(B(y, \epsilon/2))$ and let δ be a Lebesgue number. If $d(x_1, x_2) < \delta$, $\{x_1, x_2\}$ lies in one of these sets, so $d(f(x_1), f(x_2)) < \epsilon$. ■

A point x is **isolated** if $\{x\}$ is open.

Theorem 3.30 (27.7)

A nonempty compact Hausdorff space without isolated points is uncountable.

Proof.

Step 1. for any nonempty open U and point x, there is a nonempty open $V \subset U$ with $x \notin \overline{V}$ (choose $y \in U$, $y \neq x$, which exists since $U \neq \{x\}$; separate x and y by disjoint $W_1 \ni x$, $W_2 \ni y$; $V = W_2 \cap U$).

Step 2. given $f : \mathbb{Z}_+ \to X$, apply Step 1 recursively to get nonempty open $V_1 \supset V_2 \supset \cdots$ with $f(n) \notin \overline{V_n}$ and $\overline{V_{n+1}} \subset V_n$. The nested closed sets $\overline{V_n}$ have a common point x (Theorem 3.23), and $x \neq f(n)$ for all n. ■

Corollary 3.31 (27.8)

Every closed interval in $\mathbb{R}$ is uncountable.

3.6 Limit point compactness

Motivation. The cover, limit-point and sequential versions of compactness agree for metric spaces.

X is **limit point compact** if every infinite subset has a limit point, and **sequentially compact** if every sequence has a convergent subsequence.

Theorem 3.32 (28.1)

Compactness implies limit point compactness, but not conversely.

Proof. If A has no limit point, A is closed and hence compact, and each $a \in A$ has a neighbourhood U_a with $U_a \cap A = \{a\}$. The U_a cover the compact A, finitely many suffice, and each contains one point of A: A is finite. Counterexample to the converse: $X = \mathbb{Z}_+ \times Y$ with Y a two-point indiscrete space (every point has a limit point), or S_Ω below. ■

Theorem 3.33 (28.2)

For a metrizable space X the following are equivalent:
(1) compact;
(2) limit point compact;
(3) sequentially compact.

Proof. (1)⇒(2) is Theorem 3.32. (2)⇒(3): if the range of (x_n) is finite, a constant subsequence converges; otherwise it has a limit point x, and choosing $x_{n_k} \in B(x, 1/k)$ with n_k increasing (possible since each ball contains infinitely many points of the range, Theorem 2.19) gives a convergent subsequence. (3)⇒(1): first, the Lebesgue number lemma holds: if not, there are C_n of diameter $< 1/n$ in no member of $\mathcal{A}$; pick $x_n \in C_n$ and a convergent subsequence $x_{n_i} \to a$; $a \in A \in \mathcal{A}$ with $B(a, \epsilon) \subset A$, and for large i, $C_{n_i} \subset B(a, \epsilon)$, a contradiction. Second, X is **totally bounded**: if some ϵ admits no finite cover by ϵ-balls, choose $x_{n+1} \notin \bigcup_{i \le n} B(x_i, \epsilon)$; the sequence has no convergent subsequence since all distances are $\ge \epsilon$. Finally, with δ a Lebesgue number, cover X by finitely many $\delta/3$-balls; each lies in a member of $\mathcal{A}$. ■

Example 3.2

S_Ω is limit point compact but not compact: an infinite subset contains a countably infinite subset B, which has an upper bound b (Theorem 1.27), so $B \subset [a_0, b]$, which is compact (Theorem 3.24); hence B has a limit point. S_Ω is not compact since the sections S_α (or the open intervals $[a_0, \alpha)$) cover it without a finite subcover. It is also sequentially compact.

3.7 Local compactness

Motivation. Local compactness is exactly what makes the one-point compactification Hausdorff.

X is **locally compact at** x if some compact C contains a neighbourhood of x, and locally compact if this holds at every point. $\mathbb{R}$, $\mathbb{R}^n$, every simply ordered set with the least upper bound property are locally compact; $\mathbb{Q}$ and $\mathbb{R}^\omega$ are not.

Theorem 3.34 (29.1)

X is locally compact Hausdorff iff there is a space Y with:
(1) X is a subspace of Y;
(2) $Y - X$ is a single point;
(3) Y is compact Hausdorff. Two such Y are homeomorphic by a homeomorphism fixing X.

Proof. Uniqueness: map $Y \to Y'$ by the identity on X and $p \mapsto q$; open sets of Y containing p are complements of compact subsets of X ($Y - U$ is closed in Y, hence compact, and lies in X), and the same holds in Y'.

Existence: let $Y = X \cup \{\infty\}$ with open sets the open sets of X and the sets $Y - C$ for compact $C \subset X$. This is a topology (uses that compact sets in the Hausdorff X are closed); X is a subspace; Y is compact (a cover contains some $Y - C$, and C needs finitely many more); Y is Hausdorff (separate x from ∞ by $\operatorname{Int} C \ni x$ and $Y - C$, where C is a compact set containing a neighbourhood of x). Conversely, a subspace of a compact Hausdorff space with one point removed is Hausdorff and locally compact (separate x from the removed point by disjoint open $U \ni x$, $V \ni \infty$; $C = Y - V$ is compact and contains U). ■

Y is the **one-point compactification**; e.g. $\mathbb{R} \cup \{\infty\} \cong S^1$ and $\mathbb{R}^n \cup \{\infty\} \cong S^n$ (stereographic projection).

Theorem 3.35 (29.2)

A Hausdorff space X is locally compact iff for every x and neighbourhood U of x there is a neighbourhood V of x with $\overline{V}$ compact and $\overline{V} \subset U$.

Proof. ($\Leftarrow$) Take $C = \overline{V}$.

($\Rightarrow$) In the one-point compactification Y, $C = Y - U$ is closed, hence compact; by Lemma 3.18 there are disjoint open $V \ni x$, $W \supset C$. Then $\overline{V}$ (closure in Y) misses W, so $\overline{V} \subset U$; and $\overline{V}$ is compact as a closed subset of Y. ■

Corollary 3.36 (29.3)

A closed or open subspace of a locally compact Hausdorff space is locally compact.

Proof. Closed A: if C is compact containing a neighbourhood U of x, then $C \cap A$ is compact and contains $U \cap A$. Open A: by Theorem 3.35, $x \in V \subset \overline{V} \subset A$ with $\overline{V}$ compact. ■

Corollary 3.37 (29.4)

X is homeomorphic to an open subspace of a compact Hausdorff space iff X is locally compact Hausdorff.

Chapter 4. Countability and Separation Axioms

4.1 The countability axioms

Motivation. The countability axioms start the road to the metrization theorem: regular spaces with a countable basis are metrizable.

X has a **countable basis at** x if some countable collection $\mathcal{B}$ of neighbourhoods of x has every neighbourhood of x containing a member of $\mathcal{B}$; X is **first countable** if this holds at every point. Metric spaces are first countable.

Theorem 4.1 (30.1)

(a) If a sequence of points of A converges to x, then $x \in \overline{A}$; the converse holds if X is first countable.
(b) If $f : X \to Y$ is continuous, it preserves convergent sequences; the converse holds if X is first countable.

Proof. As in Lemma 2.38 and Theorem 2.39: with a countable basis $B_1, B_2, \ldots$ at x, replace it by $B_1 \cap \cdots \cap B_n$ (nested) and pick $x_n \in A \cap B_1 \cap \cdots \cap B_n$. ■

X is **second countable** if its topology has a countable basis. $\mathbb{R}^n$ (rational balls) and $\mathbb{R}^\omega$ (products of rational intervals in finitely many coordinates) are second countable; $\mathbb{R}^\omega$ in the uniform topology is first but not second countable (the sets $\{0,1\}^\omega$ give uncountably many points at mutual distance 1).

Theorem 4.2 (30.2)

Subspaces and countable products of first (resp. second) countable spaces are first (resp. second) countable.

Proof. Intersect a countable basis with the subspace. For $\prod_n X_n$ with countable bases $\mathcal{B}_n$, the products $\prod U_n$ with $U_n \in \mathcal{B}_n$ for finitely many n and $U_n = X_n$ otherwise form a countable basis (a countable union of countable sets, Theorem 1.15); similarly at a point. ■

A is **dense** if $\overline{A} = X$.

Theorem 4.3 (30.3)

If X has a countable basis, then (a) every open covering has a countable subcovering (X is **Lindelöf**), and (b) X has a countable dense subset (X is **separable**).

Proof. (a) For each basis element B_n that lies in some member of the cover, choose such a member A_n; the A_n cover X, since each x lies in some $B_n \subset A$ for A in the cover.

(b) Pick $x_n \in B_n$ (nonempty ones); every nonempty open set contains some B_n, hence some x_n. ■

Example 4.1

$\mathbb{R}_\ell$ is first countable, Lindelöf and separable but not second countable (any basis must contain, for each x, a set B_x with $x = \inf B_x$). $\mathbb{R}_\ell^2$ (the Sorgenfrey plane) is not Lindelöf: the antidiagonal $L = \{x \times (-x)\}$ is closed and each point of L has a neighbourhood $[a,b) \times [-a,d)$ meeting L only there; adding $X - L$ gives a cover with no countable subcover. So products of Lindelöf spaces need not be Lindelöf, and subspaces of Lindelöf spaces need not be Lindelöf.

For metrizable spaces, second countable, separable and Lindelöf are equivalent.

4.2 The separation axioms

Motivation. The separation axioms (Hausdorff, regular, normal) measure how far a space is from being metric.

Assume one-point sets are closed. X is **regular** if for each point x and closed set B not containing x there are disjoint open sets containing x and B; **normal** if each pair of disjoint closed sets has disjoint open neighbourhoods. Normal $\Rightarrow$ regular $\Rightarrow$ Hausdorff.

Lemma 4.4 (31.1)

(a) X is regular iff for each x and neighbourhood U of x there is a neighbourhood V of x with $\overline{V} \subset U$.
(b) X is normal iff for each closed A and open $U \supset A$ there is an open $V \supset A$ with $\overline{V} \subset U$.

Proof. (a) ($\Rightarrow$) Separate x from $B = X - U$ by disjoint $V \ni x$, $W \supset B$; then $\overline{V} \cap B = \emptyset$ since $X - W$ is closed and contains V.

($\Leftarrow$) For $x \notin B$, take $U = X - B$, then V and $X - \overline{V}$ separate.

(b) Same with A for x. ■

Theorem 4.5 (31.2)

(a) Subspaces and products of Hausdorff spaces are Hausdorff.
(b) Subspaces and products of regular spaces are regular.

Proof. (b) Subspace Y: if $x \in Y$ and B closed in Y with $x \notin B$, then $B = \overline{B} \cap Y$ and $x \notin \overline{B}$; separate x from $\overline{B}$ in X and intersect with Y. Product: given $x = (x_\alpha)$ and a basis neighbourhood $\prod U_\alpha$, choose $V_\alpha \ni x_\alpha$ with $\overline{V_\alpha} \subset U_\alpha$ (and $V_\alpha = X_\alpha$ when $U_\alpha = X_\alpha$); then $V = \prod V_\alpha$ has $\overline{V} = \prod \overline{V_\alpha} \subset \prod U_\alpha$ (Theorem 2.30). Apply Lemma 4.4. ■

Example 4.2

(a) $\mathbb{R}_K$ is Hausdorff but not regular: K is closed and $0 \notin K$, but any neighbourhood of 0 has closure meeting every open set containing K.
(b) $\mathbb{R}_\ell$ is normal, but $\mathbb{R}_\ell^2$ is regular (a product of regular spaces) and not normal (Example 4.3). So products of normal spaces need not be normal.

4.3 Normal spaces

Motivation. The main normal spaces are metric spaces, compact Hausdorff spaces, and regular spaces with a countable basis.

Theorem 4.6 (32.1)

Every regular space with a countable basis is normal.

Proof. Let $\mathcal{B}$ be a countable basis and A, B disjoint closed. Each $a \in A$ has a basis element $U \ni a$ with $\overline{U} \cap B = \emptyset$ (regularity, Lemma 4.4); enumerate those used as $U_1, U_2, \ldots$; they cover A and their closures miss B. Similarly $V_1, V_2, \ldots$ cover B with closures missing A. Put $U_n' = U_n - \bigcup_{i \le n} \overline{V_i}$, $V_n' = V_n - \bigcup_{i \le n} \overline{U_i}$, which are open. $U' = \bigcup U_n' \supset A$ and $V' = \bigcup V_n' \supset B$. If $x \in U_j' \cap V_k'$ with $j \le k$, then $x \in U_j$ but $x \notin \overline{U_j}$ (from V_k'), a contradiction; similarly for $j > k$. ∎

Theorem 4.7 (32.2)

Every metrizable space is normal.

Proof. For disjoint closed A, B choose for each $a \in A$ an ϵ_a with $B(a, \epsilon_a) \cap B = \emptyset$, and for each $b \in B$ an ϵ_b with $B(b, \epsilon_b) \cap A = \emptyset$. Let $U = \bigcup_a B(a, \epsilon_a/2)$, $V = \bigcup_b B(b, \epsilon_b/2)$. If $z \in U \cap V$ with $d(a, z) < \epsilon_a/2$, $d(b, z) < \epsilon_b/2$, then $d(a, b) < (\epsilon_a + \epsilon_b)/2 \le \max(\epsilon_a, \epsilon_b)$, contradicting the choice of radii. ∎

Theorem 4.8 (32.3)

Every compact Hausdorff space is normal.

Proof. Regularity: Lemma 3.18 with the compact closed set B. Normality: for disjoint closed A, B, choose for each $a \in A$ disjoint $U_a \ni a$, $V_a \supset B$; finitely many U_{a_i} cover A; take $U = \bigcup U_{a_i}$, $V = \bigcap V_{a_i}$. ∎

Theorem 4.9 (32.4)

Every well-ordered set is normal in the order topology.

Proof. Every interval $(x, y]$ is open (it equals (x, y^+) with y^+ the successor of y, or a ray if y is largest). Let A, B be disjoint closed, first not containing the smallest element a_0. For $a \in A$ choose $(x_a, a]$ missing B, and for $b \in B$ choose $(y_b, b]$ missing A. $U = \bigcup (x_a, a]$, $V = \bigcup (y_b, b]$ are disjoint: if $z \in (x_a, a] \cap (y_b, b]$ with $a < b$, then $y_b < z \le a < b$, so $a \in (y_b, b]$, which misses A. If $a_0 \in A$, note $\{a_0\}$ is open and closed; apply the argument to $A - \{a_0\}$ and add $\{a_0\}$ to U. ∎

Example 4.3

(a) The Sorgenfrey plane $\mathbb{R}_\ell^2$ is not normal. With L the antidiagonal, every subset $A \subset L$ is closed, as is $L - A$. If each pair A, $L - A$ had disjoint open neighbourhoods $U_A \supset A$, $V_A \supset L - A$, then $A \mapsto U_A \cap D$ (D the rational points) would be injective from $\mathcal{P}(L)$ into $\mathcal{P}(D)$, which has the cardinality of $\mathcal{P}(\mathbb{Z}_+)$; but L has the cardinality of $\mathbb{R}$, and $\mathcal{P}(\mathbb{R})$ is strictly larger (Theorem 1.18).
(b) $S_\Omega \times \overline{S}_\Omega$ is not normal although both factors are (Theorem 4.9): the diagonal $A = \{x \times x\}$ and $B = S_\Omega \times \{\Omega\}$ are disjoint closed sets that cannot be separated. Since $\overline{S}_\Omega \times \overline{S}_\Omega$ is compact Hausdorff, hence normal, normality is not hereditary.
(c) $\mathbb{R}^J$ for uncountable J is not normal.

4.4 The Urysohn lemma

Motivation. Urysohn's lemma produces, in any normal space, a continuous function that is 0 and 1 on given disjoint closed sets; it drives the metrization and extension theorems.

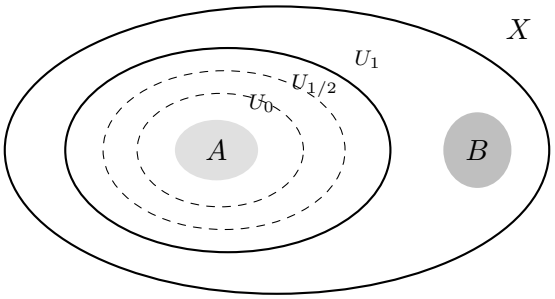


Figure. *Urysohn's construction: open sets U_r, one for each rational $r \in [0,1]$, with $A \subset U_0$, $\overline{U_r} \subset U_s$ for $r < s$, and $U_1 = X - B$. Then $f(x) = \inf\{r : x \in U_r\}$ is continuous, 0 on A and 1 on B.*

Theorem 4.10 (33.1, Urysohn lemma)

Let X be normal and A, B disjoint closed subsets. For any $[a,b] \subset \mathbb{R}$ there is a continuous $f : X \to [a,b]$ with $f = a$ on A and $f = b$ on B.

Proof. It suffices to treat $[0,1]$.

Step 1. let P be the rationals in $[0,1]$, enumerated with $1, 0$ first. Define open U_p $(p \in P)$ such that $p < q \Rightarrow \overline{U_p} \subset U_q$: put $U_1 = X - B$; by Lemma 4.4(b) choose $U_0 \supset A$ with $\overline{U_0} \subset U_1$. If U_p is defined for the first n rationals P_n with the property, let r be the next; $P_n \cup \{r\}$ is finite and ordered, r has an immediate predecessor p and successor q in it, and normality gives U_r with $\overline{U_p} \subset U_r \subset \overline{U_r} \subset U_q$. By induction (recursive definition) all U_p are defined.

Step 2. extend to all rationals p by $U_p = \emptyset$ for $p < 0$ and $U_p = X$ for $p > 1$; the property persists.

Step 3. let $\mathbb{Q}(x) = \{p : x \in U_p\}$, which contains no $p < 0$ and every $p > 1$, and $f(x) = \inf \mathbb{Q}(x) \in [0,1]$. On A, $f = 0$; on B, $x \notin U_p$ for $p \le 1$, so $f = 1$.

Step 4. (i) $x \in \overline{U_r} \Rightarrow f(x) \le r$ (then $x \in U_s$ for all rational $s > r$);

(ii) $x \notin U_r \Rightarrow f(x) \ge r$ (then $x \notin U_s$ for $s < r$).

Step 5. continuity: given x_0 and $(c,d) \ni f(x_0)$, choose rationals $c < p < f(x_0) < q < d$. Then $U = U_q - \overline{U_p}$ is an open neighbourhood of x_0 (by (i), (ii) contrapositives), and for $x \in U$: $x \in U_q \subset \overline{U_q}$ gives $f(x) \le q$; $x \notin \overline{U_p} \supset U_p$ gives $f(x) \ge p$. So $f(U) \subset [p,q] \subset (c,d)$. ∎

A, B are **separated by a continuous function** if there is $f : X \to [0,1]$ with $f(A) = \{0\}$, $f(B) = \{1\}$. A space in which one-point sets are closed and each point x_0 and closed set $A \not\ni x_0$ are so separated is **completely regular**. Normal ⇒ completely regular ⇒ regular (if $f(x_0) = 0$, $f(A) = 1$, the sets $f^{-1}([0, \frac{1}{2}))$, $f^{-1}((\frac{1}{2}, 1])$ separate).

Theorem 4.11 (33.2)

Subspaces and products of completely regular spaces are completely regular.

Proof. Subspace: restrict the function separating x_0 from $\overline{A}$. Product: given $x = (x_\alpha)$ and a basis neighbourhood $\prod U_\alpha$ with $U_\alpha \ne X_\alpha$ only for $\alpha_1, \dots, \alpha_n$, take $f_i : X_{\alpha_i} \to [0,1]$ with $f_i(x_{\alpha_i}) = 1$ and $f_i = 0$ off U_{α_i}; then $f = \prod_i f_i \circ \pi_{\alpha_i}$ is 1 at x and 0 off $\prod U_\alpha$. ∎

So $\mathbb{R}^2_\ell$ and $S_\Omega \times \overline{S}_\Omega$ are completely regular but not normal.

4.5 The Urysohn metrization theorem

Motivation. Urysohn functions and a countable basis embed the space in $\mathbb{R}^\omega$: second countable spaces are metrizable exactly when regular.

Theorem 4.12 (34.1, Urysohn metrization theorem)

Every regular space with a countable basis is metrizable.

Proof.

Step 1. there is a countable family $f_n : X \to [0,1]$ such that for each x_0 and neighbourhood U of x_0 some f_n is positive at x_0 and vanishes off U. Indeed X is normal (Theorem 4.6); for each pair (n,m) of indices with $\overline{B_n} \subset B_m$ (B_i the countable basis), Urysohn gives $g_{n,m} = 1$ on $\overline{B_n}$, 0 off B_m. Given $x_0 \in U$, choose B_m with $x_0 \in B_m \subset U$ and, by regularity, B_n with $x_0 \in B_n$, $\overline{B_n} \subset B_m$; $g_{n,m}$ works.

Step 2. by the imbedding theorem below, $F(x) = (f_1(x), f_2(x), \dots)$ imbeds X in $\mathbb{R}^\omega$ (product topology), which is metrizable (Theorem 2.36). (One may also give a direct metric $D(F(x), F(y)) = \sup|f_n(x) - f_n(y)|/n$.) ■

Theorem 4.13 (34.2, imbedding theorem)

Let X be a space in which one-point sets are closed and $\{f_\alpha\}_{\alpha \in J}$ continuous functions $X \to \mathbb{R}$ such that for each x_0 and neighbourhood U of x_0 some f_α is positive at x_0 and zero outside U. Then $F(x) = (f_\alpha(x))_{\alpha \in J}$ is an imbedding of X in $\mathbb{R}^J$. If each f_α maps into $[0,1]$, F imbeds X in $[0,1]^J$.

Proof. F is continuous (Theorem 2.31). Injective: for $x \neq y$, choose α with $f_\alpha(x) > 0$ and $f_\alpha = 0$ off $X - \{y\}$ (open since points are closed), so $f_\alpha(y) = 0$. Open onto $Z = F(X)$: let U be open in X and $z_0 = F(x_0) \in F(U)$. Choose β with $f_\beta(x_0) > 0$, $f_\beta = 0$ off U, and let $W = \pi_\beta^{-1}((0,\infty)) \cap Z$, open in Z and containing z_0. If $z = F(x) \in W$, then $f_\beta(x) > 0$, so $x \in U$ and $z \in F(U)$. Thus $F(U)$ is open in Z. ■

Corollary 4.14 (34.3)

X is completely regular iff it is homeomorphic to a subspace of $[0,1]^J$ for some J.

Proof. $[0,1]^J$ is completely regular (Theorem 4.11), hence so are its subspaces. Conversely, for completely regular X the family of all continuous $f : X \to [0,1]$ satisfies the hypothesis of Theorem 4.13 (separate x_0 from $X - U$). ■

4.6 The Tietze extension theorem

Motivation. In normal spaces a continuous function on a closed set extends, as a uniform limit of Urysohn functions.

Theorem 4.15 (35.1, Tietze extension theorem)

Let X be normal and $A \subset X$ closed.
(a) Every continuous $f : A \to [a,b]$ extends to a continuous $g : X \to [a,b]$.
(b) Every continuous $f : A \to \mathbb{R}$ extends to a continuous $g : X \to \mathbb{R}$.

Proof.

Step 1. for continuous $f : A \to [-r,r]$ there is continuous $g : X \to \mathbb{R}$ with $|g| \le r/3$ on X and $|g - f| \le 2r/3$ on A: the sets $B = f^{-1}([-r, -r/3])$, $C = f^{-1}([r/3, r])$ are disjoint and closed in X;

Urysohn gives $g : X \to [-r/3, r/3]$ with $g = -r/3$ on B and $r/3$ on C, and one checks the three cases.

Step 2. for $f : A \to [-1, 1]$ apply Step 1 repeatedly: g_1 for f with $r = 1$; g_2 for $f - g_1$ with $r = \frac{2}{3}$; in general g_n with $|g_n| \le \frac{1}{3}(\frac{2}{3})^{n-1}$ and $|f - \sum_{i \le n} g_i| \le (\frac{2}{3})^n$ on A. $g = \sum g_n$ converges uniformly (comparison with a geometric series), so g is continuous (Theorem 2.42), $|g| \le 1$, and $g = f$ on A.

Step 3. for (b), regard f as a map into $(-1, 1) \cong \mathbb{R}$. Extend it by Step 2 to $g : X \to [-1, 1]$. The set $D = g^{-1}(\{-1, 1\})$ is closed and disjoint from A; Urysohn gives $\phi : X \to [0, 1]$ with $\phi = 0$ on D, $\phi = 1$ on A. Then $h = \phi g$ extends f and maps into $(-1, 1)$. ■

Conversely, Tietze's theorem implies the Urysohn lemma (extend the function equal to 0 on A and 1 on B from the closed set $A \cup B$).

4.7 Imbeddings of manifolds

Motivation. Partitions of unity embed every compact manifold in some $\mathbb{R}^N$.

An **m-manifold** is a Hausdorff space with a countable basis in which each point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^m$. For $\phi : X \to \mathbb{R}$, the **support** is $\overline{\{\phi \neq 0\}}$. A **partition of unity dominated by** an indexed open covering $\{U_i\}$ is a family of continuous $\phi_i : X \to [0, 1]$ with $\operatorname{supp} \phi_i \subset U_i$ and $\sum \phi_i = 1$.

Theorem 4.16 (36.1, existence of finite partitions of unity)

Every finite open covering $\{U_1, \dots, U_n\}$ of a normal space has a dominated partition of unity.

Proof.

Step 1 (shrinking). there are open V_i covering X with $\overline{V_i} \subset U_i$. Inductively, $A = X - (V_1 \cup \cdots \cup V_{k-1} \cup U_{k+1} \cup \cdots \cup U_n)$ is closed and contained in U_k; choose $V_k \supset A$ with $\overline{V_k} \subset U_k$; the sets $V_1, \dots, V_k, U_{k+1}, \dots, U_n$ still cover.

Step 2. shrink again to W_i with $\overline{W_i} \subset V_i$, and take Urysohn $\psi_i = 1$ on $\overline{W_i}$, 0 off V_i; $\operatorname{supp} \psi_i \subset \overline{V_i} \subset U_i$. $\Psi = \sum \psi_i > 0$ everywhere (the W_i cover), and $\phi_i = \psi_i / \Psi$. ■

Theorem 4.17 (36.2)

Every compact m-manifold imbeds in $\mathbb{R}^N$ for some N.

Proof. Cover X by finitely many open U_i with imbeddings $g_i : U_i \to \mathbb{R}^m$, and let ϕ_i be a dominated partition of unity (X is compact Hausdorff, hence normal). Define $h_i = \phi_i g_i$ on U_i and 0 off $\operatorname{supp} \phi_i$ (continuous by the pasting lemma for open sets), and $F = (\phi_1, \dots, \phi_n, h_1, \dots, h_n) : X \to \mathbb{R}^{n(m+1)}$. F is continuous; injective: if $F(x) = F(y)$, choose i with $\phi_i(x) > 0$; then $\phi_i(y) = \phi_i(x) > 0$, so $x, y \in U_i$, and $\phi_i(x) g_i(x) = \phi_i(y) g_i(y)$ gives $g_i(x) = g_i(y)$, so $x = y$. A continuous injection of a compact space into a Hausdorff space is an imbedding (Theorem 3.20). ■

4.8 Selected exercises

Motivation. Exercises from Munkres applying the Urysohn lemma and the metrization theorem.

Exercise 4.1 (§33, Ex. 2; connected spaces are uncountable)

(a) Show that a connected normal space with more than one point is uncountable.
(b) Show that a connected regular space with more than one point is uncountable.

Solution.

(a) Let $x \neq y$ in X. One-point sets are closed, so by the Urysohn lemma (Theorem 4.10) there is a continuous $f : X \to [0,1]$ with $f(x) = 0$ and $f(y) = 1$. $f(X)$ is connected and contains 0 and 1, hence contains $[0,1]$, which is uncountable. The image of a countable set is countable, so X is uncountable.

(b) By (a) it suffices to show that a countable regular space X is normal.

Step 1. Let A, B be disjoint closed sets, both nonempty (otherwise $\emptyset$ and X separate them), and enumerate $A = \{a_1, a_2, \dots\}$, $B = \{b_1, b_2, \dots\}$ (repeating terms if a set is finite). By Lemma 4.4(a) applied to the neighbourhood $X - B$ of a_n, there is an open $U_n \ni a_n$ with $\overline{U_n} \cap B = \emptyset$; likewise an open $V_n \ni b_n$ with $\overline{V_n} \cap A = \emptyset$.

Step 2. Put $U_n' = U_n - \bigcup_{k \le n} \overline{V_k}$ and $V_n' = V_n - \bigcup_{k \le n} \overline{U_k}$, open sets. Since $\overline{V_k} \cap A = \emptyset$, $a_n \in U_n'$; so $U' = \bigcup_n U_n' \supset A$, and likewise $V' = \bigcup_n V_n' \supset B$.

Step 3. $U' \cap V' = \emptyset$: if $x \in U_j' \cap V_k'$ with $j \le k$, then $x \in U_j \subset \overline{U_j}$, but V_k' misses $\overline{U_j}$; the case $j > k$ is symmetric. So X is normal. ∎

Exercise 4.2 (§34, Ex. 3; compact metrizable spaces)

Let X be compact Hausdorff. Show that X is metrizable iff X has a countable basis.

Solution.

($\Leftarrow$) X is normal (Theorem 4.8), hence regular, and has a countable basis, so it is metrizable by Theorem 4.12.

($\Rightarrow$) Let d be a metric for X.

Step 1. For each $n \in \mathbb{Z}_+$ the balls $B(x, 1/n)$, $x \in X$, cover X; let $\mathcal{A}_n$ be a finite subcovering, and $\mathcal{B} = \bigcup_n \mathcal{A}_n$, a countable collection of open sets.

Step 2. $\mathcal{B}$ is a basis. Let U be open and $x \in U$; choose $\epsilon > 0$ with $B(x, \epsilon) \subset U$ and n with $2/n < \epsilon$. Some $B(y, 1/n) \in \mathcal{A}_n$ contains x. For $z \in B(y, 1/n)$, $d(z, x) \le d(z, y) + d(y, x) < 2/n < \epsilon$, so $x \in B(y, 1/n) \subset U$. ∎

Exercise 4.3 (§36, Ex. 4; the shrinking lemma)

A family $\{U_n\}_{n \in \mathbb{Z}_+}$ is **point-finite** if each x lies in U_n for only finitely many n. Let X be normal and $\{U_1, U_2, \dots\}$ a point-finite indexed open covering of X. Show that there is an indexed open covering $\{V_1, V_2, \dots\}$ of X with $\overline{V_n} \subset U_n$ for each n.

Solution.

Step 1. (Construction.) We choose $V_1, V_2, \dots$ inductively so that for every n,

$$\overline{V_n} \subset U_n \quad \text{and} \quad V_1 \cup \cdots \cup V_n \cup U_{n+1} \cup U_{n+2} \cup \cdots = X.$$

Given $V_1, \dots, V_{n-1}$ with this property (for $n = 1$ the hypothesis is that the U_k cover X), the set

$$A_n = X - (V_1 \cup \cdots \cup V_{n-1} \cup U_{n+1} \cup U_{n+2} \cup \cdots)$$

is closed and, by the covering property at stage $n-1$, contained in U_n. By Lemma 4.4(b) choose an open $V_n \supset A_n$ with $\overline{V_n} \subset U_n$. Then $V_1, \dots, V_n, U_{n+1}, \dots$ cover X: a point not in $V_1 \cup \cdots \cup V_{n-1} \cup U_{n+1} \cup \cdots$ lies in $A_n \subset V_n$.

Step 2. (The V_n cover X.) Let $x \in X$. By point-finiteness there is N with $x \notin U_k$ for $k > N$. By Step 1 at stage N, $x \in V_1 \cup \cdots \cup V_N \cup U_{N+1} \cup \cdots$, so $x \in V_i$ for some $i \le N$.

Without point-finiteness Step 2 fails: a point could lie in every U_k and in no V_k. For finite coverings this is Step 1 of the proof of Theorem 4.16. ∎

Chapter 5. The Tychonoff Theorem

5.1 The Tychonoff theorem

Motivation. Arbitrary products of compact spaces are compact; the proof uses Zorn's lemma, and the theorem is equivalent to the axiom of choice.

We prove compactness via the finite intersection property (Theorem 3.23): a space is compact iff every collection $\mathcal{A}$ of subsets with the finite intersection property (FIP) satisfies $\bigcap_{A\in\mathcal{A}} \overline{A} \neq \emptyset$.

Lemma 5.1 (37.1)

Let $\mathcal{A}$ be a collection of subsets of X with the FIP. Then there is a collection $\mathcal{D} \supset \mathcal{A}$ with the FIP such that no collection properly containing $\mathcal{D}$ has the FIP ($\mathcal{D}$ is *maximal* with respect to the FIP).

Proof. Let $\mathbb{A}$ be the set of all collections $\mathcal{B} \supset \mathcal{A}$ of subsets of X with the FIP, strictly partially ordered by proper inclusion. If $\mathbb{B} \subset \mathbb{A}$ is a chain, $\mathcal{C} = \bigcup_{\mathcal{B}\in\mathbb{B}} \mathcal{B}$ has the FIP: finitely many $C_1, \ldots, C_n \in \mathcal{C}$ lie in $\mathcal{B}_1, \ldots, \mathcal{B}_n$, all contained in the largest $\mathcal{B}_k$ (chain), which has the FIP. So $\mathcal{C}$ is an upper bound, and Zorn's lemma (Theorem 1.29) gives a maximal element. ■

Lemma 5.2 (37.2)

Let $\mathcal{D}$ be maximal with respect to the FIP.
(a) Every finite intersection of elements of $\mathcal{D}$ is in $\mathcal{D}$.
(b) If $A \subset X$ intersects every element of $\mathcal{D}$, then $A \in \mathcal{D}$.

Proof. (a) Let B be a finite intersection of elements of $\mathcal{D}$; $\mathcal{E} = \mathcal{D} \cup \{B\}$ has the FIP (a finite intersection of members of $\mathcal{E}$ is a finite intersection of members of $\mathcal{D}$), so $\mathcal{E} = \mathcal{D}$ by maximality.

(b) $\mathcal{E} = \mathcal{D} \cup \{A\}$: a finite intersection $A \cap D_1 \cap \cdots \cap D_n$ is nonempty since $D_1 \cap \cdots \cap D_n \in \mathcal{D}$ by (a) and A meets it. So $A \in \mathcal{D}$. ■

Theorem 5.3 (37.3, Tychonoff theorem)

An arbitrary product of compact spaces is compact in the product topology.

Proof. Let $X = \prod_{\alpha\in J} X_\alpha$ and $\mathcal{A}$ a collection of subsets of X with the FIP; extend it to a maximal $\mathcal{D}$ (Lemma 5.1). It suffices to find $x \in \bigcap_{D\in\mathcal{D}} \overline{D}$. For each α, $\{\pi_\alpha(D) : D \in \mathcal{D}\}$ has the FIP in the compact X_α, so choose $x_\alpha \in \bigcap_D \overline{\pi_\alpha(D)}$, and let $x = (x_\alpha)$.

Step 1. if U_β is a neighbourhood of x_β, then $\pi_\beta^{-1}(U_\beta)$ meets every $D \in \mathcal{D}$ (since U_β meets $\pi_\beta(D)$, as $x_\beta \in \overline{\pi_\beta(D)}$), so $\pi_\beta^{-1}(U_\beta) \in \mathcal{D}$ by Lemma 5.2(b).

Step 2. every basis element containing x is a finite intersection of such subbasis elements, hence lies in $\mathcal{D}$ (Lemma 5.2(a)), hence meets every element of $\mathcal{D}$ (FIP). So $x \in \overline{D}$ for all $D \in \mathcal{D}$. ■

The theorem fails for the box topology ($[0,1]^\omega$ in the box topology is not compact), and it is equivalent to the axiom of choice.

Example 5.1

$[0,1]^\omega$ is a compact metrizable space; every regular second countable space imbeds in it (proof of Theorem 4.12). $[0,1]^J$ for uncountable J is compact Hausdorff but not metrizable: it is

not even first countable (a countable family of basis neighbourhoods of a point restricts only countably many coordinates).

5.2 The Stone–Čech compactification

Motivation. Every completely regular space has a largest compactification βX, the closure of X embedded in a product of intervals, to which every bounded continuous function extends.

A **compactification** of X is a compact Hausdorff space Y containing X as a subspace with $\overline{X} = Y$; two are **equivalent** if there is a homeomorphism between them that is the identity on X. If X has a compactification, it is completely regular (a subspace of a compact Hausdorff, hence normal, space).

Lemma 5.4 (38.1)

If $h : X \to Z$ is an imbedding in a compact Hausdorff space Z, there is a compactification Y of X and an imbedding $H : Y \to Z$ equal to h on X; Y is unique up to equivalence (Y is the compactification **induced** by h).

Proof. Let $X_0 = h(X)$ and $Y_0 = \overline{X_0}$, a compact Hausdorff space. Form $Y = X \sqcup (Y_0 - X_0)$ with the bijection $H : Y \to Y_0$ (h on X, identity elsewhere) and give Y the topology making H a homeomorphism. Then X is a subspace of Y, dense since X_0 is dense in Y_0. ■

Theorem 5.5 (38.2)

Every completely regular space X has a compactification Y such that every bounded continuous $f : X \to \mathbb{R}$ extends uniquely to a continuous $Y \to \mathbb{R}$.

Proof. Let $\{f_\alpha\}_{\alpha \in J}$ be all bounded continuous real functions on X, with $f_\alpha(X) \subset I_\alpha$, where $I_\alpha = [\inf f_\alpha, \sup f_\alpha]$. $h = (f_\alpha) : X \to Z = \prod I_\alpha$ is an imbedding (Theorem 4.13, using complete regularity), and Z is compact Hausdorff (Tychonoff). Let Y be the induced compactification with $H : Y \to Z$. For $f = f_\beta$, $\pi_\beta \circ H$ is a continuous extension. Uniqueness follows from the next lemma. ■

Lemma 5.6 (38.3)

If $A \subset X$ and $f : A \to Z$ is continuous with Z Hausdorff, there is at most one continuous extension $g : \overline{A} \to Z$.

Proof. If $g \neq g'$ at x, separate $g(x), g'(x)$ by disjoint U, U'; the neighbourhood $g^{-1}(U) \cap g'^{-1}(U')$ of x contains a point $y \in A$, but $g(y) = g'(y) = f(y)$ lies in $U \cap U'$. ■

Theorem 5.7 (38.4)

If Y is a compactification of the completely regular X with the extension property of Theorem 5.5, every continuous map f of X into a compact Hausdorff space C extends uniquely to a continuous $g : Y \to C$.

Proof. C is completely regular, so imbeds in some $[0,1]^J$ (Corollary 4.14); regard $C \subset [0,1]^J$. Each coordinate $\pi_\alpha \circ f$ extends to $g_\alpha : Y \to [0,1]$; $g = (g_\alpha) : Y \to [0,1]^J$ is continuous. Since $g(X) = f(X) \subset C$ and X is dense, $g(Y) \subset \overline{g(X)} \subset C$ (C is closed).

Uniqueness: Lemma 5.6. ■

Theorem 5.8 (38.5)

Two compactifications Y_1, Y_2 of X with the extension property of Theorem 5.5 are equivalent.

Proof. Extend the inclusion $j_2 : X \to Y_2$ to $f_2 : Y_1 \to Y_2$ and j_1 to $f_1 : Y_2 \to Y_1$ (Theorem 5.7). $f_1 \circ f_2 : Y_1 \to Y_1$ extends the identity of X, as does id_{Y_1}; by Lemma 5.6, $f_1 \circ f_2 = \mathrm{id}$. Similarly $f_2 \circ f_1 = \mathrm{id}$. ■

This compactification is the **Stone–Čech compactification** $\beta(X)$. It is characterized by the universal property of Theorem 5.7.

Example 5.2

(a) The one-point compactification of $(0,1)$ is the circle and $[0,1]$ is a two-point compactification; neither is $\beta(0,1)$, since $\sin(1/x)$ extends to $\beta(0,1)$.
(b) $\beta(\mathbb{Z}_+)$ is not sequentially compact: $\mathbb{Z}_+$ has no convergent subsequence in $\beta\mathbb{Z}_+$ (separate even and odd subsequences by extending functions on $\mathbb{Z}_+$).
(c) $\beta(S_\Omega) = \overline{S}_\Omega$, because every continuous real function on S_Ω is eventually constant.

5.3 Selected exercises

Motivation. Exercises from Munkres on the countable intersection criterion, maximality of $\beta(X)$, the fact about S_Ω used in Example 5.2(c), and connectedness of βX.

Exercise 5.1 (§37, Ex. 2; Lindelöf spaces)

A collection $\mathcal{A}$ of subsets of X has the **countable intersection property** if every countable intersection of elements of $\mathcal{A}$ is nonempty. Show that X is Lindelöf iff for every collection $\mathcal{A}$ with the countable intersection property, $\bigcap_{A\in\mathcal{A}} \overline{A} \neq \emptyset$.

Solution.

($\Rightarrow$) Let X be Lindelöf, $\mathcal{A}$ with the countable intersection property, and suppose $\bigcap_{A\in\mathcal{A}} \overline{A} = \emptyset$. Then $\{X - \overline{A} : A \in \mathcal{A}\}$ is an open covering of X, so countably many $X - \overline{A_n}$ cover X, i.e. $\bigcap_n \overline{A_n} = \emptyset$. But $\bigcap_n A_n \subset \bigcap_n \overline{A_n}$ is nonempty, a contradiction.

($\Leftarrow$) Let $\mathcal{U}$ be an open covering with no countable subcovering, and $\mathcal{A} = \{X - U : U \in \mathcal{U}\}$. For $U_1, U_2, \cdots \in \mathcal{U}$, $\bigcap_n (X - U_n) = X - \bigcup_n U_n \neq \emptyset$, so $\mathcal{A}$ has the countable intersection property. Its members are closed, and $\bigcap_{U\in\mathcal{U}}(X - U) = X - \bigcup \mathcal{U} = \emptyset$, contradicting the hypothesis. So X is Lindelöf. ■

Exercise 5.2 (§38, Ex. 4; $\beta(X)$ is the largest compactification)

Let Y be any compactification of the completely regular space X. Show that there is a continuous, closed, surjective map $g : \beta(X) \to Y$ that is the identity on X.

Solution.

Step 1. The inclusion $j : X \to Y$ is a continuous map into a compact Hausdorff space, so by Theorem 5.7 it extends to a continuous $g : \beta(X) \to Y$, and $g = j$ is the identity on X.

Step 2. (g is closed.) A closed $C \subset \beta(X)$ is compact (Theorem 3.16), so $g(C)$ is compact (Theorem 3.19), hence closed in the Hausdorff space Y (Theorem 3.17).

Step 3. (g is onto.) By Step 2, $g(\beta(X))$ is closed in Y; it contains X, which is dense in Y. So $g(\beta(X)) = Y$.

A closed continuous surjection is a quotient map, so every compactification of X is equivalent to a quotient space of $\beta(X)$. ■

Exercise 5.3 (§38, Ex. 5(a); functions on S_Ω)

Show that every continuous $f : S_\Omega \to \mathbb{R}$ is eventually constant: there is $\alpha \in S_\Omega$ with $f(\beta) = f(\alpha)$ for all $\beta > \alpha$.

Solution.

Step 1. (Increasing sequences converge.) Let $\alpha_1 < \alpha_2 < \cdots$ in S_Ω. By Theorem 1.27 the set of upper bounds of $\{\alpha_n\}$ in S_Ω is nonempty; let b be its least element. If $c < b$, then c is not an upper bound, so $\alpha_N > c$ for some N, and $\alpha_n \in (c, b]$ for $n \geq N$. Since $b > \alpha_1$ is not the least element of S_Ω, every neighbourhood of b contains such an interval $(c, b]$ with $c < b$; hence $\alpha_n \to b$, and by continuity $f(\alpha_n) \to f(b)$.

Step 2. (Small oscillation eventually.) Fix $\epsilon > 0$. We claim there is α with $|f(\beta) - f(\alpha)| < \epsilon$ for all $\beta > \alpha$. If not, then for every α there is $\beta > \alpha$ with $|f(\beta) - f(\alpha)| \geq \epsilon$. Starting from any α_1, we obtain $\alpha_1 < \alpha_2 < \cdots$ with $|f(\alpha_{n+1}) - f(\alpha_n)| \geq \epsilon$ for all n. Then $(f(\alpha_n))$ is not Cauchy, contradicting Step 1.

Step 3. (Conclusion.) For each n choose α_n as in Step 2 with $\epsilon = 1/n$, and by Theorem 1.27 an upper bound α of $\{\alpha_n\}$. For $\beta > \alpha$ and each n, both β and α are $\geq \alpha_n$, so

$$|f(\beta) - f(\alpha)| \leq |f(\beta) - f(\alpha_n)| + |f(\alpha_n) - f(\alpha)| < \tfrac{2}{n}$$

(the second term is 0 if $\alpha = \alpha_n$). Letting $n \to \infty$, $f(\beta) = f(\alpha)$. ■

Exercise 5.4 (§38, Ex. 6; connectedness of $\beta(X)$)

Let X be completely regular. Show that X is connected iff $\beta(X)$ is connected.

Solution.

($\Rightarrow$) X is connected and dense in $\beta(X)$, and the closure of a connected set is connected, so $\beta(X) = \overline{X}$ is connected.

($\Leftarrow$) Suppose $X = A \cup B$ is a separation. Then $f = 0$ on A, $f = 1$ on B is continuous and bounded (A, B are open), so it extends to a continuous $g : \beta(X) \to \mathbb{R}$ (Theorem 5.5). Since X is dense, $g(\beta(X)) = g(\overline{X}) \subset \overline{g(X)} = \{0, 1\}$, and both values occur. So $g^{-1}(0)$ and $g^{-1}(1)$ form a separation of $\beta(X)$, a contradiction. ■

Chapter 6. Metrization Theorems and Paracompactness

6.1 Local finiteness

Motivation. Locally finite families replace the countable basis, since piecewise definitions over them behave well.

A collection $\mathcal{A}$ of subsets of X is **locally finite** if every point has a neighbourhood meeting only finitely many elements of $\mathcal{A}$; it is **countably locally finite** (σ-locally finite) if it is a countable union of locally finite collections. A collection $\mathcal{B}$ **refines** $\mathcal{A}$ (is a **refinement** of $\mathcal{A}$) if each $B \in \mathcal{B}$ lies in some $A \in \mathcal{A}$; it is an open (closed) refinement if its elements are open (closed).

Lemma 6.1 (39.1)

Let $\mathcal{A}$ be locally finite.
(a) Every subcollection of $\mathcal{A}$ is locally finite.
(b) $\{\overline{A} : A \in \mathcal{A}\}$ is locally finite.
(c) $\overline{\bigcup_{A\in\mathcal{A}} A} = \bigcup_{A\in\mathcal{A}} \overline{A}$.

Proof. (a) is clear.

(b) An open set U meeting $\overline{A}$ meets A; so a neighbourhood meeting finitely many A meets only the corresponding $\overline{A}$.

(c) The inclusion $\supset$ always holds. Let $x \in \overline{\bigcup A}$ and let U be a neighbourhood of x meeting only $A_1, \dots, A_k$. If $x \notin \overline{A_1} \cup \cdots \cup \overline{A_k}$, then $U - (\overline{A_1} \cup \cdots \cup \overline{A_k})$ is a neighbourhood of x missing every element of $\mathcal{A}$, a contradiction. ■

Without local finiteness (c) fails: the one-point sets $\{1/n\}$ have union with closure containing 0.

Lemma 6.2 (39.2)

If X is metrizable, every open covering $\mathcal{A}$ of X has a countably locally finite open refinement $\mathcal{E}$ that covers X.

Proof. Fix a metric d and a well-ordering $<$ of $\mathcal{A}$ (Theorem 1.25). For $n \in \mathbb{Z}_+$ and $U \in \mathcal{A}$ let

$$S_n(U) = \{x : B(x, 1/n) \subset U\}, \qquad T_n(U) = S_n(U) - \bigcup_{V<U} V, \qquad E_n(U) = \bigcup_{x\in T_n(U)} B(x, 1/3n).$$

If $V < U$, $x \in T_n(V)$, $y \in T_n(U)$, then $B(x, 1/n) \subset V$ and $y \notin V$, so $d(x, y) \geq 1/n$. Hence distinct sets $E_n(U), E_n(V)$ are at distance at least $1/3n$, a ball of radius $1/6n$ meets at most one of them, and $\mathcal{E}_n = \{E_n(U)\}_U$ is locally finite. Also $E_n(U) \subset U$, since $B(x, 1/3n) \subset B(x, 1/n) \subset U$ for $x \in T_n(U)$. Finally, given x, let U be the first element of $\mathcal{A}$ containing x and choose n with $B(x, 1/n) \subset U$: then $x \in S_n(U)$ and x lies in no $V < U$, so $x \in T_n(U) \subset E_n(U)$. Take $\mathcal{E} = \bigcup_n \mathcal{E}_n$. ■

6.2 The Nagata–Smirnov metrization theorem

Motivation. A space is metrizable if and only if it is regular with a countably locally finite basis.

A set is a G_δ **set** if it is a countable intersection of open sets. In a metric space every closed set A is G_δ: $A = \bigcap_n \{x : d(x, A) < 1/n\}$.

Lemma 6.3 (40.1)

Let X be regular with a countably locally finite basis $\mathcal{B}$. Then X is normal and every closed set is a G_δ set.

Proof.

Step 1. for each open W there are open $U_1, U_2, \ldots$ with $W = \bigcup U_n = \bigcup \overline{U_n}$. Write $\mathcal{B} = \bigcup \mathcal{B}_n$ with $\mathcal{B}_n$ locally finite, let $\mathcal{C}_n = \{B \in \mathcal{B}_n : \overline{B} \subset W\}$ (locally finite) and $U_n = \bigcup_{B \in \mathcal{C}_n} B$. By Lemma 6.1(c), $\overline{U_n} = \bigcup_{\mathcal{C}_n} \overline{B} \subset W$. If $x \in W$, regularity gives a basis element $B \ni x$ with $\overline{B} \subset W$; $B \in \mathcal{C}_n$ for some n, so $x \in U_n$.

Step 2. if C is closed, apply Step 1 to $W = X - C$: $C = \bigcap_n (X - \overline{U_n})$, a G_δ.

Step 3. for disjoint closed C, D, Step 1 with $W = X - D$ gives open U_n covering C with closures missing D, and with $W = X - C$ open V_n covering D with closures missing C. Now finish exactly as in the proof of Theorem 4.6. ■

Lemma 6.4 (40.2)

If X is normal and A is a closed G_δ set, there is a continuous $f : X \to [0, 1]$ with $f = 0$ on A and $f > 0$ off A.

Proof. Write $A = \bigcap U_n$ with U_n open. Urysohn gives $f_n : X \to [0, 1]$ equal to 0 on A and 1 on $X - U_n$. Let $f = \sum_n f_n / 2^n$, continuous by uniform convergence (Theorem 2.42). It vanishes on A; if $x \notin A$, then $x \notin U_n$ for some n, so $f(x) \geq 2^{-n} > 0$. ■

Theorem 6.5 (40.3, Nagata–Smirnov metrization theorem)

X is metrizable iff X is regular and has a countably locally finite basis.

Proof. ($\Leftarrow$)

Step 1. let $\mathcal{B} = \bigcup \mathcal{B}_n$ with $\mathcal{B}_n$ locally finite. X is normal and $X - B$ is a closed G_δ for each $B \in \mathcal{B}$ (Lemma 6.3), so Lemma 6.4 gives $f_{n,B} : X \to [0, 1/n]$ positive on B and zero off B. For $x_0 \in U$ choose $B \in \mathcal{B}_n$ with $x_0 \in B \subset U$; then $f_{n,B}(x_0) > 0$ and $f_{n,B} = 0$ off U.

Step 2. let $J = \{(n, B) : B \in \mathcal{B}_n\}$ and $F(x) = (f_{n,B}(x)) \in [0, 1]^J$; give $[0, 1]^J$ the uniform metric $\bar{\rho}$. By the proof of Theorem 4.13, F is injective and maps open sets of X to sets open in $F(X)$ for the product topology, hence also for the finer uniform topology (Theorem 2.35).

Step 3 (F is continuous into the uniform metric). let $x_0 \in X$ and $\epsilon > 0$. For each n choose a neighbourhood U_n of x_0 meeting only finitely many elements of $\mathcal{B}_n$; all but finitely many $f_{n,B}$ vanish on U_n, and for the remaining ones continuity gives a smaller neighbourhood W_n on which $|f_{n,B}(x) - f_{n,B}(x_0)| \leq \epsilon/2$ for every $B \in \mathcal{B}_n$. Choose N with $1/N \leq \epsilon/2$ and let $W = W_1 \cap \cdots \cap W_N$. For $x \in W$ the coordinates with $n \leq N$ differ by at most $\epsilon/2$, and those with $n > N$ lie in $[0, 1/n]$, so differ by at most $1/n < \epsilon/2$. Thus $\bar{\rho}(F(x), F(x_0)) \leq \epsilon/2 < \epsilon$, and F imbeds X in a metric space.

($\Rightarrow$) A metrizable space is regular (Theorem 4.7). For each m let $\mathcal{A}_m$ be the covering by balls of radius $1/m$ and $\mathcal{D}_m$ a countably locally finite open refinement covering X (Lemma 6.2); its elements have diameter at most $2/m$. $\mathcal{D} = \bigcup_m \mathcal{D}_m$ is countably locally finite. It is a basis: if $x \in U$ open, choose m with $B(x, 1/m) \subset U$ and $D \in \mathcal{D}_{3m}$ containing x; D has diameter at most $2/3m < 1/m$, so $D \subset B(x, 1/m) \subset U$. ■

6.3 Paracompactness

Motivation. Paracompactness, every open cover has a locally finite refinement, gives partitions of unity for arbitrary covers, and metric spaces have it.

X is **paracompact** if every open covering has a locally finite open refinement that covers X. Compact spaces are paracompact, and so is $\mathbb{R}^n$ (Theorem 6.9 below).

Theorem 6.6 (41.1)

Every paracompact Hausdorff space is normal.

Proof. Regularity: let $a \notin B$, B closed. For each $b \in B$ choose open $U_b \ni b$ with $a \notin \overline{U_b}$ (Hausdorff). Cover X by the U_b and $X - B$, take a locally finite open refinement $\mathcal{C}$ covering X, and let $\mathcal{D}$ be the elements of $\mathcal{C}$ meeting B. Each $D \in \mathcal{D}$ is not inside $X - B$, so $D \subset U_b$ for some b and $a \notin \overline{D}$. $V = \bigcup_{\mathcal{D}} D$ is open, contains B, and $\overline{V} = \bigcup \overline{D} \not\ni a$ (Lemma 6.1). Then V and $X - \overline{V}$ separate B and a. Normality: the same argument with a closed set A in place of a, using regularity to choose U_b with $\overline{U_b} \cap A = \emptyset$. ∎

Theorem 6.7 (41.2)

Every closed subspace of a paracompact space is paracompact.

Proof. Let Y be closed and $\mathcal{A}$ a covering of Y by sets open in Y; for each $A \in \mathcal{A}$ choose A' open in X with $A' \cap Y = A$. Cover X by the A' together with $X - Y$, and take a locally finite open refinement $\mathcal{B}$ covering X. Then $\{B \cap Y : B \in \mathcal{B}\}$ covers Y, is locally finite, consists of sets open in Y, and refines $\mathcal{A}$ (a B inside $X - Y$ gives the empty set). ∎

A paracompact space can have non-paracompact subspaces: $\overline{S}_\Omega$ is compact, but S_Ω is not paracompact (it is not metrizable yet is locally metrizable; see Theorem 6.14). The product $\mathbb{R}^2_\ell$ of paracompact spaces is not paracompact, because it is Hausdorff but not normal.

Lemma 6.8 (41.3, Michael)

Let X be regular. The following are equivalent: every open covering of X has a refinement that is (1) an open covering of X and countably locally finite;
(2) a covering of X and locally finite;
(3) a closed covering of X and locally finite;
(4) an open covering of X and locally finite.

Proof. (4)⇒(1) is trivial.

(1)⇒(2): let $\mathcal{B} = \bigcup \mathcal{B}_n$ be an open refinement as in (1), $V_i = \bigcup_{U \in \mathcal{B}_i} U$, and for $U \in \mathcal{B}_n$ put $S_n(U) = U - \bigcup_{i<n} V_i$. The sets $S_n(U)$ refine $\mathcal{A}$ and cover X: for x, take the smallest N with x in some $U \in \mathcal{B}_N$; then $x \in S_N(U)$. Local finiteness: with $x \in U \in \mathcal{B}_N$, the open set $U \subset V_N$ meets no $S_n(V)$ with $n > N$; for each $n \le N$ choose a neighbourhood W_n meeting finitely many elements of $\mathcal{B}_n$, hence finitely many $S_n(V)$. Then $U \cap W_1 \cap \cdots \cap W_N$ meets finitely many sets $S_n(V)$.

(2)⇒(3): let $\mathcal{B}$ be the open sets U with $\overline{U}$ contained in some element of $\mathcal{A}$; it covers X by regularity. Take a locally finite refinement $\mathcal{C}$ covering X; $\{\overline{C}\}$ is locally finite (Lemma 6.1), closed, covers X and refines $\mathcal{A}$.

(3)⇒(4): let $\mathcal{B}$ be a locally finite refinement of $\mathcal{A}$ covering X (by (3)). Let $\mathcal{C}$ be the open sets meeting only finitely many elements of $\mathcal{B}$; it covers X. By (3) take a closed locally finite refinement $\mathcal{D}$ of $\mathcal{C}$ covering X; each $D \in \mathcal{D}$ meets only finitely many elements of $\mathcal{B}$. For $B \in \mathcal{B}$ let $C(B) =$

$X - \bigcup\{D \in \mathcal{D} : D \cap B = \emptyset\}$, open by Lemma 6.1(c) and containing B, and choose $F(B) \in \mathcal{A}$ with $B \subset F(B)$. The collection $\mathcal{E} = \{C(B) \cap F(B) : B \in \mathcal{B}\}$ is an open refinement of $\mathcal{A}$ covering X. It is locally finite: a point has a neighbourhood W meeting only $D_1, \ldots, D_k \in \mathcal{D}$, so $W \subset D_1 \cup \cdots \cup D_k$. If D meets $C(B)$, then D meets B (otherwise $D \subset X - C(B)$); hence each D_i meets only finitely many $C(B)$, and so does W. ■

Theorem 6.9 (41.4)

Every metrizable space is paracompact.

Proof. It is regular, and condition (1) of Lemma 6.8 holds by Lemma 6.2. ■

Theorem 6.10 (41.5)

Every regular Lindelöf space is paracompact.

Proof. A countable subcovering is a countable union of one-element (hence locally finite) collections, so condition (1) of Lemma 6.8 holds. ■

So $\mathbb{R}_\ell$ is paracompact, while $\mathbb{R}_\ell^2$ is not; and $\mathbb{R}^\omega$ in the product topology is paracompact (metrizable), whereas $\mathbb{R}^J$ for uncountable J is not (it is not normal).

Lemma 6.11 (41.6)

Let X be paracompact Hausdorff and $\{U_\alpha\}_{\alpha \in J}$ an indexed open covering. There is a locally finite indexed open covering $\{V_\alpha\}_{\alpha \in J}$ with $\overline{V_\alpha} \subset U_\alpha$ for each α (a **precise refinement**).

Proof. Let $\mathcal{A}$ be the open sets A with $\overline{A}$ in some U_α; it covers X by regularity (Theorem 6.6). Take a locally finite open refinement $\mathcal{B}$ covering X, and for each B choose $f(B) \in J$ with $\overline{B} \subset U_{f(B)}$. Let $V_\alpha = \bigcup\{B : f(B) = \alpha\}$. The V_α are open and cover X; a neighbourhood meeting finitely many B meets finitely many V_α; and $\overline{V_\alpha} = \bigcup_{f(B)=\alpha} \overline{B} \subset U_\alpha$ by Lemma 6.1(c). ■

Theorem 6.12 (41.7)

If X is paracompact Hausdorff and $\{U_\alpha\}_{\alpha \in J}$ is an indexed open covering, there is a partition of unity dominated by $\{U_\alpha\}$: continuous $\phi_\alpha : X \to [0,1]$ with $\operatorname{supp} \phi_\alpha \subset U_\alpha$, $\{\operatorname{supp} \phi_\alpha\}$ locally finite, and $\sum_\alpha \phi_\alpha = 1$.

Proof. By Lemma 6.11 twice, take locally finite open coverings $\{V_\alpha\}$, $\{W_\alpha\}$ with $\overline{W_\alpha} \subset V_\alpha$, $\overline{V_\alpha} \subset U_\alpha$. Urysohn (with X normal) gives ψ_α equal to 1 on $\overline{W_\alpha}$ and 0 off V_α, so $\operatorname{supp} \psi_\alpha \subset \overline{V_\alpha} \subset U_\alpha$, and these supports are locally finite (Lemma 6.1(b)). $\Psi = \sum \psi_\alpha$ is near each point a finite sum of continuous functions, hence continuous, and positive since the W_α cover. Put $\phi_\alpha = \psi_\alpha / \Psi$. ■

Theorem 6.13 (41.8)

Let X be paracompact Hausdorff, $\mathcal{C}$ a locally finite collection of subsets, and $\epsilon_C > 0$ for each $C \in \mathcal{C}$. There is a continuous $f : X \to \mathbb{R}$ with $f > 0$ everywhere and $f(x) \le \epsilon_C$ for $x \in C$.

Proof. Cover X by open sets each meeting finitely many elements of $\mathcal{C}$, and take a dominated partition of unity $\{\phi_U\}$ indexed by this covering. Let δ_U be the minimum of ϵ_C over the finitely many C meeting U ($\delta_U = 1$ if there are none), and $f = \sum_U \delta_U \phi_U$, continuous and positive. If $x \in C$ and $\phi_U(x) > 0$, then $x \in U$, so U meets C and $\delta_U \le \epsilon_C$; hence $f(x) \le \epsilon_C \sum \phi_U(x) = \epsilon_C$. ■

6.4 The Smirnov metrization theorem

Motivation. A paracompact Hausdorff space is metrizable exactly when it is locally metrizable.

X is **locally metrizable** if each point has a neighbourhood that is metrizable in the subspace topology.

Theorem 6.14 (42.1, Smirnov metrization theorem)

X is metrizable iff it is a paracompact Hausdorff space that is locally metrizable.

Proof. ($\Rightarrow$) Theorem 6.9.

($\Leftarrow$) X is regular (Theorem 6.6); we find a countably locally finite basis and apply Theorem 6.5. Cover X by open metrizable sets and take a locally finite open refinement $\mathcal{C}$ covering X; each $C \in \mathcal{C}$ has a metric d_C, and the balls $B_C(x,\epsilon) = \{y \in C : d_C(x,y) < \epsilon\}$ are open in X. For each m, let $\mathcal{A}_m$ be the covering by all balls $B_C(x, 1/m)$ ($C \in \mathcal{C}$, $x \in C$), and $\mathcal{D}_m$ a locally finite open refinement covering X. Then $\mathcal{D} = \bigcup \mathcal{D}_m$ is countably locally finite. Basis: let $x \in U$ open. Only finitely many $C_1, \dots, C_k \in \mathcal{C}$ contain x; choose ϵ_i with $B_{C_i}(x,\epsilon_i) \subset U \cap C_i$ and m with $2/m < \min \epsilon_i$. Choose $D \in \mathcal{D}_m$ with $x \in D$; then $D \subset B_C(y, 1/m)$ for some C and y, and $x \in C$, so $C = C_i$ for some i. By the triangle inequality $D \subset B_{C_i}(x, 2/m) \subset U$. ■

6.5 Selected exercises

Motivation. Exercises from Munkres on why countable local finiteness is needed, a product theorem for paracompactness, and non-paracompactness of S_Ω.

Exercise 6.1 (§39, Ex. 5; countably locally finite means countable)

Let X have a countable basis. Show that a collection $\mathcal{A}$ of subsets of X is countably locally finite iff it is countable.

Solution.

($\Leftarrow$) A countable $\mathcal{A} = \{A_1, A_2, \dots\}$ is the countable union of the collections $\{A_n\}$, each of which is finite, hence locally finite.

($\Rightarrow$) It suffices to show that a locally finite collection $\mathcal{A}_0$ is countable. Let $\mathcal{B}$ be a countable basis.

Step 1. Let $\mathcal{B}' \subset \mathcal{B}$ consist of the basis elements meeting only finitely many elements of $\mathcal{A}_0$. $\mathcal{B}'$ covers X: each x has a neighbourhood W meeting finitely many elements of $\mathcal{A}_0$, and a basis element B with $x \in B \subset W$ lies in $\mathcal{B}'$.

Step 2. Every nonempty $A \in \mathcal{A}_0$ contains a point, which lies in some $B \in \mathcal{B}'$; so A meets B. Hence $\mathcal{A}_0 - \{\emptyset\} \subset \bigcup_{B \in \mathcal{B}'} \{A \in \mathcal{A}_0 : A \cap B \neq \emptyset\}$, a countable union of finite sets, and $\mathcal{A}_0$ is countable (Theorem 1.15). ■

Exercise 6.2 (§40, Ex. 3; locally finite bases)

Show that a T_1 space X with a locally finite basis $\mathcal{B}$ is discrete.

Solution. Let $x \in X$.

Step 1. Some neighbourhood W of x meets only finitely many elements of $\mathcal{B}$. Every basis element containing x meets W, so there are only finitely many, say $B_1, \dots, B_k$ ($k \geq 1$ since $\mathcal{B}$ covers X). Then $O = B_1 \cap \cdots \cap B_k$ is an open set containing x.

Step 2. $O = \{x\}$. Let $y \neq x$. Since $\{y\}$ is closed, $X - \{y\}$ is a neighbourhood of x, so some $B \in \mathcal{B}$ has $x \in B \subset X - \{y\}$. This B is one of the B_i, so $y \notin O$.

Hence every one-point set is open and X is discrete. ■

Exercise 6.3 (§41, Ex. 2; products with compact spaces)

(a) Show that if X is paracompact and Y is compact, then $X \times Y$ is paracompact.
(b) Conclude that S_Ω is not paracompact.

Solution.

(a) Let $\mathcal{A}$ be an open covering of $X \times Y$.

Step 1. For $x \in X$, the compact slice $x \times Y$ is covered by finitely many $A_{x,1}, \ldots, A_{x,k_x} \in \mathcal{A}$. By the tube lemma (Lemma 3.21) there is an open $W_x \ni x$ with $W_x \times Y \subset A_{x,1} \cup \cdots \cup A_{x,k_x}$.

Step 2. $\{W_x\}$ covers X; let $\mathcal{C}$ be a locally finite open refinement covering X, and for each $C \in \mathcal{C}$ choose $x(C)$ with $C \subset W_{x(C)}$. Let

$$\mathcal{D} = \{(C \times Y) \cap A_{x(C),i} : C \in \mathcal{C},\ 1 \leq i \leq k_{x(C)}\}.$$

Its elements are open and each lies in an element of $\mathcal{A}$.

Step 3. $\mathcal{D}$ covers $X \times Y$: if $x \in C \in \mathcal{C}$, then $x \times Y \subset C \times Y \subset W_{x(C)} \times Y \subset \bigcup_i A_{x(C),i}$, so each point of $x \times Y$ lies in some $(C \times Y) \cap A_{x(C),i}$.

Step 4. $\mathcal{D}$ is locally finite: for (x, y), choose a neighbourhood N of x meeting only finitely many $C \in \mathcal{C}$. Then $N \times Y$ meets only the elements of $\mathcal{D}$ coming from these C, finitely many for each.

(b) $\overline{S}_\Omega$ is compact: it is a closed interval of a well-ordered set, which has the least upper bound property (the set of upper bounds of a bounded nonempty subset has a least element), so Theorem 3.24 applies. If S_Ω were paracompact, then $S_\Omega \times \overline{S}_\Omega$ would be paracompact by (a), and Hausdorff, hence normal by Theorem 6.6. This contradicts Example 4.3(b). ■

Chapter 7. Complete Metric Spaces and Function Spaces

7.1 Complete metric spaces

Motivation. Completeness depends on the metric, but function spaces with the uniform metric are complete and every metric space can be completed.

A sequence (x_n) in (X,d) is **Cauchy** if for every $\epsilon>0$ there is N with $d(x_n,x_m)<\epsilon$ for $n,m\geq N$; (X,d) is **complete** if every Cauchy sequence converges. A closed subset of a complete space is complete, and a complete subspace of a metric space is closed. (X,d) is complete iff it is complete in the standard bounded metric $\bar{d}=\min(d,1)$, since the two have the same Cauchy sequences.

Lemma 7.1 (43.1)

A metric space is complete if every Cauchy sequence has a convergent subsequence.

Proof. Let (x_n) be Cauchy with $x_{n_i}\to x$. Given ϵ, choose N with $d(x_n,x_m)<\epsilon/2$ for $n,m\geq N$, and i with $n_i\geq N$ and $d(x_{n_i},x)<\epsilon/2$. Then $d(x_n,x)<\epsilon$ for $n\geq N$. ■

Theorem 7.2 (43.2)

$\mathbb{R}^k$ is complete in the euclidean metric d and in the square metric ρ.

Proof. Since $\rho\leq d\leq\sqrt{k}\rho$, the two metrics have the same Cauchy sequences; use ρ. A Cauchy sequence is bounded, so it lies in a cube $[-M,M]^k$, which is compact (Theorem 3.26), hence sequentially compact (Theorem 3.33). Apply Lemma 7.1. ■

Lemma 7.3 (43.3)

In $X=\prod X_\alpha$ (product topology), $x_n\to x$ iff $\pi_\alpha(x_n)\to\pi_\alpha(x)$ for each α.

Proof. ($\Rightarrow$) Each π_α is continuous.

($\Leftarrow$) A basis element $\prod U_\alpha$ containing x restricts only $\alpha_1,\ldots,\alpha_k$; choose N_i with $\pi_{\alpha_i}(x_n)\in U_{\alpha_i}$ for $n\geq N_i$, and let $N=\max N_i$. ■

Theorem 7.4 (43.4)

There is a metric on $\mathbb{R}^\omega$ (product topology) in which it is complete.

Proof. Take $D(x,y)=\sup_i \bar{d}(x_i,y_i)/i$, which induces the product topology (Theorem 2.36). If (x^n) is D-Cauchy, then $\bar{d}(x_i^n,x_i^m)\leq iD(x^n,x^m)$, so each coordinate sequence is Cauchy in $\mathbb{R}$ and converges. By Lemma 7.3, (x^n) converges in $\mathbb{R}^\omega$. ■

For a metric space (Y,d) and a set J, the **uniform metric** on Y^J is $\bar{\rho}(x,y)=\sup_\alpha \bar{d}(x_\alpha,y_\alpha)$.

Theorem 7.5 (43.5)

If Y is complete in d, then Y^J is complete in the uniform metric $\bar{\rho}$.

Proof. Y is complete in $\bar{d}$. If (f_n) is $\bar{\rho}$-Cauchy, then $\bar{d}(f_n(\alpha), f_m(\alpha)) \le \bar{\rho}(f_n, f_m)$, so each $f_n(\alpha)$ converges, say to $f(\alpha)$. Given ϵ, choose N with $\bar{\rho}(f_n, f_m) < \epsilon/2$ for $n, m \ge N$. For fixed $n \ge N$ and α, letting $m \to \infty$ gives $\bar{d}(f_n(\alpha), f(\alpha)) \le \epsilon/2$; hence $\bar{\rho}(f_n, f) \le \epsilon/2 < \epsilon$. ■

Let $\mathcal{C}(X, Y)$ be the continuous and $\mathcal{B}(X, Y)$ the bounded functions $X \to Y$. On $\mathcal{B}(X, Y)$ the **sup metric** is $\rho(f, g) = \sup_x d(f(x), g(x))$; there $\bar{\rho} = \min(\rho, 1)$.

Theorem 7.6 (43.6)

Let X be a space and (Y, d) a metric space. $\mathcal{C}(X, Y)$ and $\mathcal{B}(X, Y)$ are closed in Y^X under the uniform metric. Hence if Y is complete, they are complete in $\bar{\rho}$, and $\mathcal{B}(X, Y)$ is complete in the sup metric ρ.

Proof. If $f_n \to f$ in $\bar{\rho}$ with f_n continuous, then $f_n \to f$ uniformly (for $\epsilon < 1$, $\bar{\rho} < \epsilon$ means $d < \epsilon$ everywhere), so f is continuous (Theorem 2.42). If the f_n are bounded, choose N with $\bar{\rho}(f_N, f) < \frac{1}{2}$; then $d(f_N(x), f(x)) < \frac{1}{2}$ for all x, so $\operatorname{diam} f(X) \le \operatorname{diam} f_N(X) + 1$. Completeness follows from Theorem 7.5, and for ρ because ρ and $\bar{\rho} = \min(\rho, 1)$ have the same Cauchy and convergent sequences. ■

Theorem 7.7 (43.7)

Every metric space (X, d) has an isometric imbedding into a complete metric space.

Proof. $\mathcal{B}(X, \mathbb{R})$ is complete in the sup metric (Theorem 7.6). Fix x_0 and for $a \in X$ define $\phi_a(x) = d(x, a) - d(x, x_0)$; it is bounded by $d(a, x_0)$. Now $|\phi_a(x) - \phi_b(x)| = |d(x, a) - d(x, b)| \le d(a, b)$, with equality at $x = a$; so $\rho(\phi_a, \phi_b) = d(a, b)$ and $a \mapsto \phi_a$ is an isometric imbedding. ■

The closure of the image is a **completion** of X: a complete metric space containing an isometric copy of X as a dense subset. It is unique up to an isometry fixing X. Completeness is not a topological property: $\mathbb{R}$ is complete but the homeomorphic $(-1, 1)$ is not.

7.2 A space-filling curve

Motivation. A uniform limit of polygonal paths gives a continuous surjection of $[0, 1]$ onto the square.

Theorem 7.8 (44.1)

Let $I = [0, 1]$. There is a continuous map $f : I \to I^2$ whose image is all of I^2.

Proof.

Step 1. call a path in a square $[a, a + s] \times [b, b + s]$ *triangular* if it runs linearly from the lower left corner to the centre of the top edge and then to the lower right corner. The operation: divide the square into four subsquares of side $s/2$ and replace the path by four triangular paths, one per subsquare, in the order lower left, upper left, upper right, lower right (oriented to keep the original endpoints); on a path made of several triangular paths, apply it to each.

Step 2. let f_0 be the triangular path in I^2 and f_n the result of applying the operation n times. Then I is divided into 4^n intervals of length 4^{-n}, and on each such interval f_n is a triangular path in a subsquare of side 2^{-n}; the 4^n subsquares are exactly the squares of the grid of side 2^{-n}. On the same interval f_{n+1} stays in the same subsquare. Hence $d(f_n(t), f_{n+1}(t)) \le \sqrt{2}\, 2^{-n}$ for all t, and

(f_n) is Cauchy in the sup metric on $\mathcal{C}(I, I^2)$, which is complete (Theorem 7.6, I^2 being closed in the complete $\mathbb{R}^2$). Let f be the limit; it is continuous, and $d(f(t), f_n(t)) \le \sum_{k\ge n} \sqrt{2}\,2^{-k} = 2\sqrt{2}\,2^{-n}$.

Step 3. given $x \in I^2$, x lies in some grid square of side 2^{-n}, which f_n visits, so there is t_n with $d(f_n(t_n), x) \le \sqrt{2}\,2^{-n}$. Then $d(f(t_n), x) \le 3\sqrt{2}\,2^{-n} \to 0$, so $x \in \overline{f(I)} = f(I)$, the image being compact and hence closed. ■

7.3 Compactness in metric spaces

Motivation. A metric space is compact iff complete and totally bounded; this finds compact subsets of function spaces.

(X, d) is **totally bounded** if for each $\epsilon > 0$ it is covered by finitely many ϵ-balls.

Theorem 7.9 (45.1)

A metric space is compact iff it is complete and totally bounded.

Proof. ($\Rightarrow$) A Cauchy sequence has a convergent subsequence (Theorem 3.33), hence converges (Lemma 7.1); and finitely many of the ϵ-balls cover.

($\Leftarrow$) We show sequential compactness. Let (x_n) be given. Cover X by finitely many balls of radius 1; one, B_1, contains x_n for n in an infinite set J_1. Given an infinite J_{k-1}, cover X by finitely many balls of radius $1/k$; one, B_k, contains x_n for infinitely many $n \in J_{k-1}$; call them J_k. Choose $n_1 < n_2 < \cdots$ with $n_k \in J_k$. For $i, j \ge k$, $x_{n_i}, x_{n_j} \in B_k$, so $d(x_{n_i}, x_{n_j}) < 2/k$. The subsequence is Cauchy and converges. ■

A family $\mathcal{F} \subset \mathcal{C}(X, Y)$ is **equicontinuous at** x_0 if for each $\epsilon > 0$ there is a neighbourhood U of x_0 with $d(f(x), f(x_0)) < \epsilon$ for all $x \in U$ and all $f \in \mathcal{F}$; it is **equicontinuous** if this holds at every point.

Lemma 7.10 (45.2)

Let X be a space and (Y, d) a metric space. If $\mathcal{F} \subset \mathcal{C}(X, Y)$ is totally bounded under the uniform metric, then $\mathcal{F}$ is equicontinuous under d.

Proof. Let $0 < \epsilon < 1$ and $x_0 \in X$. Cover $\mathcal{F}$ by $\bar\rho$-balls $B(f_1, \epsilon/3), \dots, B(f_n, \epsilon/3)$, and choose a neighbourhood U of x_0 with $d(f_i(x), f_i(x_0)) < \epsilon/3$ for all i and $x \in U$. For $f \in \mathcal{F}$ pick i with $\bar\rho(f, f_i) < \epsilon/3$; since $\epsilon/3 < 1$, $d(f(x), f_i(x)) < \epsilon/3$ for all x. For $x \in U$ the triangle inequality gives $d(f(x), f(x_0)) < \epsilon$. ■

Lemma 7.11 (45.3)

Let X and Y be compact, (Y, d) metric. If $\mathcal{F} \subset \mathcal{C}(X, Y)$ is equicontinuous under d, then $\mathcal{F}$ is totally bounded under the sup metric and the uniform metric.

Proof. Since $\bar\rho \le \rho$, it suffices to treat ρ. Let $\epsilon > 0$. For each $a \in X$ choose a neighbourhood U_a with $d(f(x), f(a)) < \epsilon/3$ for $x \in U_a$, $f \in \mathcal{F}$; finitely many $U_{a_1}, \dots, U_{a_k}$ cover X. Cover Y by finitely many open $V_1, \dots, V_m$ of diameter less than $\epsilon/3$. For each function $\alpha : \{1, \dots, k\} \to \{1, \dots, m\}$ for which some $f \in \mathcal{F}$ has $f(a_i) \in V_{\alpha(i)}$ for all i, choose one such f_α. Given $f \in \mathcal{F}$, take α with $f(a_i) \in V_{\alpha(i)}$ for all i. For $x \in U_{a_i}$,

$$d(f(x), f_\alpha(x)) \le d(f(x), f(a_i)) + d(f(a_i), f_\alpha(a_i)) + d(f_\alpha(a_i), f_\alpha(x)) < \epsilon.$$

The continuous function $d(f(x), f_\alpha(x))$ attains its maximum on the compact X, so $\rho(f, f_\alpha) < \epsilon$. The finitely many balls $B_\rho(f_\alpha, \epsilon)$ cover $\mathcal{F}$. ■

$\mathcal{F} \subset \mathcal{C}(X, \mathbb{R}^n)$ is **pointwise bounded** if $\mathcal{F}_a = \{f(a) : f \in \mathcal{F}\}$ is bounded for each a.

Theorem 7.12 (45.4, Ascoli's theorem, classical version)

Let X be compact and give $\mathcal{C}(X, \mathbb{R}^n)$ the uniform topology (d the square or the euclidean metric). A subspace $\mathcal{F}$ has compact closure iff $\mathcal{F}$ is equicontinuous and pointwise bounded under d.

Proof. Let $\mathcal{G} = \overline{\mathcal{F}}$; on $\mathcal{C}(X, \mathbb{R}^n)$ the sup metric ρ induces the uniform topology.

($\Rightarrow$) $\mathcal{G}$ is compact, hence totally bounded under ρ, and so is $\mathcal{F}$; by Lemma 7.10, $\mathcal{F}$ is equicontinuous. Evaluation $f \mapsto f(a)$ is continuous, so $\mathcal{G}_a$ is compact, hence bounded, and $\mathcal{F}_a \subset \mathcal{G}_a$.

($\Leftarrow$)

Step 1. $\mathcal{G}$ is equicontinuous and pointwise bounded. Given a and ϵ, choose U with $d(f(x), f(a)) < \epsilon/3$ for $x \in U$, $f \in \mathcal{F}$; for $g \in \mathcal{G}$ choose $f \in \mathcal{F}$ with $\rho(f, g) < \epsilon/3$; then $d(g(x), g(a)) < \epsilon$ on U. Also each $g(a)$ is within 1 of $\mathcal{F}_a$, so $\mathcal{G}_a$ is bounded.

Step 2. the union of the images $g(X)$, $g \in \mathcal{G}$, lies in a compact $Y \subset \mathbb{R}^n$. Choose neighbourhoods U_a with $d(g(x), g(a)) < 1$ for $x \in U_a$, $g \in \mathcal{G}$, and cover X by $U_{a_1}, \ldots, U_{a_k}$. Every $g(x)$ is within 1 of the bounded set $\mathcal{G}_{a_1} \cup \cdots \cup \mathcal{G}_{a_k}$, so all images lie in a closed bounded cube Y, compact by Theorem 3.26.

Step 3. $\mathcal{G} \subset \mathcal{C}(X, Y)$ is closed in the complete space $\mathcal{C}(X, \mathbb{R}^n)$ (Theorems 7.2, 7.6), hence complete; it is totally bounded by Lemma 7.11; hence compact by Theorem 7.9. ■

Corollary 7.13 (45.5)

Let X be compact. A subspace $\mathcal{F}$ of $\mathcal{C}(X, \mathbb{R}^n)$ (uniform topology) is compact iff it is closed, bounded under the sup metric ρ, and equicontinuous under d.

Proof. A compact $\mathcal{F}$ is closed and bounded in the metric space, and equicontinuous by Theorem 7.12. Conversely, if $\mathcal{F}$ is closed it equals its closure; ρ-boundedness implies pointwise boundedness; apply Theorem 7.12. ■

Example 7.1

Let $f_n : [0, 1] \to \mathbb{R}$ be differentiable with $|f_n(0)| \leq 1$ and $|f_n'| \leq 1$. By the mean value theorem $|f_n(x) - f_n(y)| \leq |x - y|$ (equicontinuity) and $|f_n| \leq 2$ (pointwise boundedness), so some subsequence converges uniformly. The family $\{x^n\}$ on $[0, 1]$ is bounded but not equicontinuous at 1, and indeed has no uniformly convergent subsequence.

7.4 Pointwise and compact convergence

Motivation. Compact convergence, between uniform and pointwise, is the topology on function spaces that preserves continuity.

For a set X and space Y, the **topology of pointwise convergence** on Y^X has subbasis $S(x, U) = \{f : f(x) \in U\}$ ($x \in X$, U open in Y); it is the product topology.

Theorem 7.14 (46.1)

$f_n \to f$ in the topology of pointwise convergence iff $f_n(x) \to f(x)$ for each $x \in X$.

Proof. This is Lemma 7.3 for the product topology. ■

Let X be a space and (Y, d) a metric space. For $f \in Y^X$, compact $C \subset X$ and $\epsilon > 0$ let $B_C(f, \epsilon) = \{g : \sup_{x \in C} d(f(x), g(x)) < \epsilon\}$. These sets form a basis: if $g \in B_C(f, \epsilon)$ and $\delta = \epsilon - \sup_C d(f, g)$, then $B_C(g, \delta) \subset B_C(f, \epsilon)$, and if g lies in $B_{C_1}(f_1, \epsilon_1) \cap B_{C_2}(f_2, \epsilon_2)$ then, with the corresponding δ_1, δ_2, $B_{C_1 \cup C_2}(g, \min \delta_i)$ lies in both. The resulting topology is the **topology of compact convergence**.

Theorem 7.15 (46.2)

$f_n \to f$ in the topology of compact convergence iff for each compact $C \subset X$, $f_n|C \to f|C$ uniformly.

Proof. Every basis element containing f contains some $B_C(f, \epsilon)$, so convergence means: for every C and ϵ, eventually $\sup_C d(f_n, f) < \epsilon$. That is uniform convergence on each C. ■

X is **compactly generated** if $A \subset X$ is open whenever $A \cap C$ is open in C for every compact C (equivalently for closed sets).

Lemma 7.16 (46.3)

If X is locally compact, or first countable, then X is compactly generated.

Proof. Suppose $A \cap C$ is open in C for every compact C, and $x \in A$. If X is locally compact, take a compact C containing a neighbourhood U of x; then $A \cap U$ is open in U, hence in X, and $x \in A \cap U \subset A$. If X is first countable and A is not open, some $x \in A$ has no neighbourhood inside A; with a nested countable basis $B_1 \supset B_2 \supset \cdots$ at x, pick $x_n \in B_n - A$. Then $x_n \to x$, and $C = \{x\} \cup \{x_n\}$ is compact. The set $A \cap C$ is open in C and contains x, so it contains x_n for large n, a contradiction. ■

Lemma 7.17 (46.4)

If X is compactly generated, then $f : X \to Y$ is continuous whenever $f|C$ is continuous for every compact $C \subset X$.

Proof. For open $V \subset Y$, $f^{-1}(V) \cap C = (f|C)^{-1}(V)$ is open in C for every C, so $f^{-1}(V)$ is open. ■

Theorem 7.18 (46.5)

Let X be compactly generated and (Y, d) metric. Then $\mathcal{C}(X, Y)$ is closed in Y^X in the topology of compact convergence.

Proof. Let f be in the closure of $\mathcal{C}(X, Y)$ and C compact. For each n choose $f_n \in B_C(f, 1/n) \cap \mathcal{C}(X, Y)$. Then $f_n|C \to f|C$ uniformly, so $f|C$ is continuous (Theorem 2.42). By Lemma 7.17, f is continuous. ■

Corollary 7.19 (46.6)

If X is compactly generated, (Y, d) metric, and continuous f_n converge to f in the topology of compact convergence, then f is continuous.

Proof. The limit of a sequence in a set lies in its closure; apply Theorem 7.18. ■

Theorem 7.20 (46.7)

For Y^X with (Y,d) metric: (uniform) $\supset$ (compact convergence) $\supset$ (pointwise convergence). If X is compact the first two coincide; if X is discrete the last two coincide.

Proof. If $g \in B_C(f,\epsilon)$ and $\delta = \epsilon - \sup_C d(f,g)$, the uniform ball $B_{\bar\rho}(g, \min(\delta,1))$ lies in $B_C(f,\epsilon)$, because $\bar\rho(g,h) < \min(\delta,1)$ forces $d(g(x),h(x)) < \delta$ for all x. If $g \in S(x,U)$, choose ϵ with $B(g(x),\epsilon) \subset U$; then $B_{\{x\}}(g,\epsilon) \subset S(x,U)$. If X is compact, for $\epsilon \le 1$ the uniform ball $B_{\bar\rho}(f,\epsilon)$ is exactly $B_X(f,\epsilon)$. If X is discrete, compact sets are finite and $B_C(f,\epsilon) = \bigcap_{x\in C} S(x, B(f(x),\epsilon))$. ■

For spaces X, Y, the **compact-open topology** on $\mathcal{C}(X,Y)$ has subbasis $S(C,U) = \{f : f(C) \subset U\}$ with C compact and U open.

Theorem 7.21 (46.8)

If (Y,d) is metric, the compact-open topology on $\mathcal{C}(X,Y)$ equals the topology of compact convergence.

Proof. Let $f \in S(C,U)$. The continuous function $y \mapsto d(y, Y-U)$ is positive on the compact $f(C)$ (if $U \neq Y$), so has a positive minimum ϵ; then $B_C(f,\epsilon) \subset S(C,U)$. Conversely consider a basic set $B_C(f,\epsilon)$ with f continuous (it suffices to treat those centred at f). For each $x \in C$ choose a neighbourhood V_x with $f(V_x) \subset B(f(x),\epsilon/4)$, so $f(\overline{V_x}) \subset B(f(x),\epsilon/3)$. Cover C by $V_{x_1},\dots,V_{x_n}$ and put $C_i = \overline{V_{x_i}} \cap C$ (compact), $U_i = B(f(x_i),\epsilon/3)$. Then $f \in W = \bigcap S(C_i,U_i)$. If $g \in W$ and $x \in C_i$, both $g(x)$ and $f(x)$ lie in U_i, so $d(f(x),g(x)) < 2\epsilon/3$. Hence $\sup_C d(f,g) \le 2\epsilon/3 < \epsilon$ and $W \subset B_C(f,\epsilon)$. ■

Corollary 7.22 (46.9)

For metric Y, the topology of compact convergence on $\mathcal{C}(X,Y)$ does not depend on the metric of Y; hence if X is compact, neither does the uniform topology on $\mathcal{C}(X,Y)$.

Proof. The compact-open topology is defined without a metric; use Theorems 7.21 and 7.20. ■

Theorem 7.23 (46.10)

If X is locally compact Hausdorff and $\mathcal{C}(X,Y)$ has the compact-open topology, the **evaluation map** $e : X \times \mathcal{C}(X,Y) \to Y$, $e(x,f) = f(x)$, is continuous.

Proof. Let $f(x) \in V$, V open. By Theorem 3.35 there is a neighbourhood U of x with $\overline{U}$ compact and $\overline{U} \subset f^{-1}(V)$. Then $U \times S(\overline{U}, V)$ is a neighbourhood of (x,f) that e maps into V. ■

Theorem 7.24 (46.11)

Give $\mathcal{C}(X,Y)$ the compact-open topology. If $f : X \times Z \to Y$ is continuous, so is the induced map $F : Z \to \mathcal{C}(X,Y)$, $F(z)(x) = f(x,z)$. The converse holds if X is locally compact Hausdorff.

Proof. Each $F(z)$ is continuous, as a composite $x \mapsto (x,z) \mapsto f(x,z)$. Let $F(z_0) \in S(C,U)$; then $C \times \{z_0\} \subset f^{-1}(U)$, and since C is compact the tube lemma (Lemma 3.21) gives a neighbourhood W of z_0 with $C \times W \subset f^{-1}(U)$; thus $F(W) \subset S(C,U)$. Conversely, $f = e \circ (\mathrm{id}_X \times F)$ is continuous by Theorem 7.23. ■

Thus, for locally compact Hausdorff X, a homotopy $X \times I \to Y$ is the same as a path in $\mathcal{C}(X, Y)$.

7.5 Ascoli's theorem

Motivation. Ascoli's theorem: equicontinuous pointwise bounded families have uniformly convergent subsequences.

Theorem 7.25 (47.1, Ascoli's theorem)

Let X be a space, (Y, d) a metric space, and $\mathcal{C}(X, Y)$ with the topology of compact convergence; let $\mathcal{F} \subset \mathcal{C}(X, Y)$.
(a) If $\mathcal{F}$ is equicontinuous under d and each $\mathcal{F}_a = \{f(a) : f \in \mathcal{F}\}$ has compact closure, then $\mathcal{F}$ lies in a compact subspace of $\mathcal{C}(X, Y)$.
(b) The converse holds if X is locally compact Hausdorff.

Proof. (a) Let $\mathcal{G}$ be the closure of $\mathcal{F}$ in Y^X with the product topology.

Step 1. $\mathcal{G}$ is equicontinuous and $\mathcal{G}_a \subset \overline{\mathcal{F}_a}$. Given a and ϵ, take U with $d(f(x), f(a)) < \epsilon/3$ for $x \in U$, $f \in \mathcal{F}$. For $g \in \mathcal{G}$ and $x \in U$, the product neighbourhood $\{h : h(x) \in B(g(x), \epsilon/3),\ h(a) \in B(g(a), \epsilon/3)\}$ of g contains some $f \in \mathcal{F}$, whence $d(g(x), g(a)) < \epsilon$. In particular $\mathcal{G} \subset \mathcal{C}(X, Y)$. Since π_a is continuous, $\pi_a(\mathcal{G}) \subset \overline{\pi_a(\mathcal{F})}$.

Step 2. $\mathcal{G}$ is compact in the product topology, being closed in $\prod_a \overline{\mathcal{F}_a}$, which is compact by Tychonoff (Theorem 5.3).

Step 3. on $\mathcal{G}$ the product topology and the topology of compact convergence agree. One is coarser (Theorem 7.20); for the other, let $g \in \mathcal{G}$, C compact, $\epsilon > 0$. For $x \in C$ choose U_x with $d(h(y), h(x)) < \epsilon/3$ for $y \in U_x$, $h \in \mathcal{G}$; cover C by $U_{x_1}, \dots, U_{x_n}$, and let $W = \{h \in \mathcal{G} : d(h(x_i), g(x_i)) < \epsilon/3\ \forall i\}$, open in the product topology. For $h \in W$ and $y \in C \cap U_{x_i}$, $d(h(y), g(y)) < \epsilon$ by three applications of the triangle inequality; the maximum over compact C is $< \epsilon$, so $W \subset B_C(g, \epsilon)$. Therefore $\mathcal{G}$ is a compact subspace of $\mathcal{C}(X, Y)$ containing $\mathcal{F}$.

(b) Let $\mathcal{H} \supset \mathcal{F}$ be compact in $\mathcal{C}(X, Y)$. Evaluation at a is continuous (the pointwise topology is coarser), so $\mathcal{H}_a$ is compact and $\overline{\mathcal{F}_a} \subset \mathcal{H}_a$ is compact. For equicontinuity at a, choose a compact C containing a neighbourhood of a. Restriction $r : \mathcal{C}(X, Y) \to \mathcal{C}(C, Y)$ is continuous when $\mathcal{C}(C, Y)$ has the uniform topology, since $r^{-1}(B_C(r(f), \epsilon)) \supset B_C(f, \epsilon)$ and for compact C these balls give the uniform topology. So $r(\mathcal{H})$ is compact, hence totally bounded under $\bar{\rho}$, hence equicontinuous on C (Lemma 7.10). As C contains a neighbourhood of a, $\mathcal{H}$, and therefore $\mathcal{F}$, is equicontinuous at a. ■

For $Y = \mathbb{R}^n$, the condition on $\mathcal{F}_a$ is pointwise boundedness, and for compact X the theorem reduces to Theorem 7.12.

7.6 Selected exercises

Motivation. Exercises from Munkres: extension of uniformly continuous maps, the contraction principle, and Arzelà's theorem.

Exercise 7.1 (§43, Ex. 2; extension of uniformly continuous maps)

Let (X, d_X) and (Y, d_Y) be metric spaces with Y complete, $A \subset X$, and $f : A \to Y$ uniformly continuous. Show that f extends uniquely to a continuous $g : \overline{A} \to Y$, and that g is uniformly continuous.

Solution.

Step 1. (Images of convergent sequences are Cauchy.) Let $a_n \in A$, $a_n \to x$. Given $\epsilon > 0$, choose $\delta > 0$ with $d_Y(f(a), f(a')) < \epsilon$ whenever $a, a' \in A$, $d_X(a, a') < \delta$. Since (a_n) converges it is Cauchy, so $d_X(a_n, a_m) < \delta$ and hence $d_Y(f(a_n), f(a_m)) < \epsilon$ for n, m large. As Y is complete, $f(a_n)$ converges.

Step 2. (Definition of g.) For $x \in \overline{A}$ there are $a_n \in A$ with $a_n \to x$ (Lemma 2.38); put $g(x) = \lim f(a_n)$. This does not depend on the sequence: if also $a'_n \to x$, the sequence $a_1, a'_1, a_2, a'_2, \ldots$ converges to x, so by Step 1 its image converges, and the subsequences $f(a_n)$, $f(a'_n)$ have the same limit. Using constant sequences, $g = f$ on A.

Step 3. (g is uniformly continuous.) Given $\epsilon > 0$, choose δ as in Step 1 for $\epsilon/2$. Let $x, y \in \overline{A}$ with $d_X(x, y) < \delta$, and $a_n \to x$, $b_n \to y$ in A. Since $d_X(a_n, b_n) \to d_X(x, y) < \delta$, for large n we have $d_X(a_n, b_n) < \delta$ and $d_Y(f(a_n), f(b_n)) < \epsilon/2$. Letting $n \to \infty$ (the metric is continuous), $d_Y(g(x), g(y)) \le \epsilon/2 < \epsilon$.

Step 4. (Uniqueness.) Two continuous extensions agree on A, hence on $\overline{A}$ by Lemma 5.6. ■

Exercise 7.2 (§43, Ex. 5; the contraction principle)

Let (X, d) be a nonempty complete metric space and $f : X \to X$ a **contraction**: there is $\alpha < 1$ with $d(f(x), f(y)) \le \alpha\, d(x, y)$ for all x, y. Show that f has a unique fixed point.

Solution.

Step 1. Fix $x_0 \in X$ and let $x_n = f^n(x_0)$. By induction $d(x_{k+1}, x_k) \le \alpha^k d(x_1, x_0)$. For $m > n$,

$$d(x_m, x_n) \le \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \le \sum_{k=n}^{\infty} \alpha^k d(x_1, x_0) = \frac{\alpha^n}{1-\alpha}\, d(x_1, x_0),$$

which tends to 0 as $n \to \infty$. So (x_n) is Cauchy and converges to some $x \in X$.

Step 2. f is continuous (given ϵ, take $\delta = \epsilon$), so $f(x) = \lim f(x_n) = \lim x_{n+1} = x$.

Step 3. If also $f(y) = y$, then $d(x, y) = d(f(x), f(y)) \le \alpha\, d(x, y)$, so $(1 - \alpha) d(x, y) \le 0$ and $x = y$. ■

Exercise 7.3 (§45, Ex. 3; Arzelà's theorem)

Let X be compact and $f_n \in \mathcal{C}(X, \mathbb{R}^k)$. Show that if $\{f_n\}$ is pointwise bounded and equicontinuous, then (f_n) has a uniformly convergent subsequence.

Solution.

Step 1. Let $\mathcal{F} = \{f_n\}$. By Ascoli's theorem (Theorem 7.12), the closure $\overline{\mathcal{F}}$ in $\mathcal{C}(X, \mathbb{R}^k)$ with the uniform topology is compact.

Step 2. The uniform topology comes from the uniform metric, so $\overline{\mathcal{F}}$ is a compact metrizable space, hence sequentially compact (Theorem 3.33).

Step 3. Therefore the sequence (f_n) in $\overline{\mathcal{F}}$ has a subsequence converging in the uniform metric, i.e. uniformly on X (for the bounded metric $\bar{d}$, uniform convergence with respect to $\bar{d}$ and to d are the same, since $\bar{d} = \min(d, 1)$). ■

Chapter 8. Baire Spaces and Dimension Theory

8.1 Baire spaces

Motivation. The Baire category theorem, a complete metric space is not a countable union of nowhere-dense sets, proves existence by showing the "bad" set is small.

$A \subset X$ has **empty interior** if it contains no nonempty open set. X is a **Baire space** if every countable union of closed sets with empty interior has empty interior.

Example 8.1

$\mathbb{Q}$ is not a Baire space: it is the countable union of its one-point sets, which are closed with empty interior. $\mathbb{Z}_+$ is a Baire space: every nonempty subset is open, so the only set with empty interior is $\emptyset$. The irrationals form a Baire space, since they admit a complete metric (they are homeomorphic to $\mathbb{Z}_+^\omega$; not proved here).

Lemma 8.1 (48.1)

X is a Baire space iff for every countable collection $\{U_n\}$ of dense open sets, $\bigcap U_n$ is dense.

Proof. U is open and dense iff $X - U$ is closed with empty interior, and $\bigcap U_n$ is dense iff $X - \bigcap U_n = \bigcup (X - U_n)$ has empty interior. ■

Theorem 8.2 (48.2, Baire category theorem)

If X is a compact Hausdorff space or a complete metric space, then X is a Baire space.

Proof. Let A_n be closed with empty interior and U_0 nonempty open; we find a point of U_0 outside every A_n. Since $U_0 \not\subset A_1$, pick $y \in U_0 - A_1$. X is regular (Theorem 4.8, or metric), so there is a neighbourhood U_1 of y with $\overline{U_1} \cap A_1 = \emptyset$ and $\overline{U_1} \subset U_0$; in the metric case choose also $\operatorname{diam} U_1 < 1$. Inductively, given nonempty open U_{n-1}, choose nonempty open U_n with $\overline{U_n} \subset U_{n-1}$, $\overline{U_n} \cap A_n = \emptyset$, and (metric case) $\operatorname{diam} U_n < 1/n$. The nested nonempty closed sets $\overline{U_n}$ have nonempty intersection: by the finite intersection property in the compact case, and by Lemma 8.3 in the complete case. A point of the intersection lies in U_0 and in no A_n. ■

Lemma 8.3 (48.3)

If $C_1 \supset C_2 \supset \cdots$ are nonempty closed sets in a complete metric space with $\operatorname{diam} C_n \to 0$, then $\bigcap C_n \neq \emptyset$.

Proof. Pick $x_n \in C_n$. For $n, m \geq N$, both lie in C_N, so $d(x_n, x_m) \leq \operatorname{diam} C_N$; the sequence is Cauchy and converges to some x. For each k, $x_n \in C_k$ for $n \geq k$, so $x \in C_k$. ■

Lemma 8.4 (48.4)

An open subspace Y of a Baire space X is a Baire space.

Proof. Let A_n be closed in Y with empty interior in Y, and $\overline{A_n}$ its closure in X, so $\overline{A_n} \cap Y = A_n$. If U is a nonempty open subset of X contained in $\overline{A_n}$, then U meets $A_n \subset Y$, and $U \cap Y$ is a nonempty open subset of Y contained in A_n, impossible. So $\overline{A_n}$ has empty interior in X. If U is open in Y (hence in X) and $U \subset \bigcup A_n \subset \bigcup \overline{A_n}$, then $U = \emptyset$ because X is Baire. ■

Theorem 8.5 (48.5)

Let X be a Baire space, (Y,d) metric, and $f_n : X \to Y$ continuous with $f_n(x) \to f(x)$ for every x. Then the set of points where f is continuous is dense in X.

Proof. For $N \in \mathbb{Z}_+$ and $\epsilon > 0$ let $A_N(\epsilon) = \{x : d(f_n(x), f_m(x)) \le \epsilon \ \forall n, m \ge N\}$, closed as an intersection of closed sets.

Step 1. every nonempty open U contains a nonempty open V such that, for some N, $d(f_n(x), f(x)) \le \epsilon$ for all $x \in V$, $n \ge N$. Indeed, since $(f_n(x))$ is Cauchy for each x, $X = \bigcup_N A_N(\epsilon)$; U is Baire (Lemma 8.4), so some $A_N(\epsilon) \cap U$ has nonempty interior V in U, and V is open in X. Letting $m \to \infty$ in the defining inequality gives the claim.

Step 2. let G_M be the union of all open V as in Step 1 for $\epsilon = 1/M$. It is open and, by Step 1, dense. So $C = \bigcap_M G_M$ is dense (Lemma 8.1). Let $x_0 \in C$ and $\epsilon > 0$; choose M with $1/M < \epsilon/3$, an open $V \ni x_0$ and N with $d(f_n(x), f(x)) \le 1/M$ on V for $n \ge N$, and a neighbourhood $W \subset V$ of x_0 with $d(f_N(x), f_N(x_0)) < \epsilon/3$ on W. Then for $x \in W$,

$$d(f(x), f(x_0)) \le d(f(x), f_N(x)) + d(f_N(x), f_N(x_0)) + d(f_N(x_0), f(x_0)) < \epsilon.$$

So f is continuous at each point of the dense set C. ∎

8.2 A nowhere-differentiable function

Motivation. By Baire's theorem most continuous functions are nowhere differentiable.

Theorem 8.6 (49.1)

Let $h : [0,1] \to \mathbb{R}$ be continuous and $\epsilon > 0$. There is a continuous, nowhere differentiable $g : [0,1] \to \mathbb{R}$ with $|h(x) - g(x)| < \epsilon$ for all x.

Proof. Let $I = [0,1]$ and $\mathcal{C} = \mathcal{C}(I, \mathbb{R})$ with the sup metric ρ; it is complete (Theorem 7.6), hence Baire. For $f \in \mathcal{C}$, $x \in I$ and $0 < k \le \frac{1}{2}$ let

$$\Delta f(x,k) = \max\Big\{\Big|\frac{f(x+k) - f(x)}{k}\Big|, \Big|\frac{f(x-k) - f(x)}{k}\Big|\Big\},$$

the maximum taken over those of $x \pm k$ lying in I (at least one does), and $\|\Delta_k f\| = \inf_x \Delta f(x,k)$. Let U_n be the set of f for which $\|\Delta_k f\| > n$ for some $k \le 1/n$.

(a) $\bigcap U_n$ consists of nowhere differentiable functions. If $f \in \bigcap U_n$, there are $k_n \le 1/n$ with $\Delta f(x, k_n) > n$ for all x. If f were differentiable at x (one-sidedly at an endpoint), the difference quotients would tend to $f'(x)$, so $\Delta f(x, k_n) \to |f'(x)|$, contradicting $\Delta f(x, k_n) > n$.

(b) U_n is open. Let $f \in U_n$ with $M = \|\Delta_k f\| > n$, and $\delta = (M - n)k/4$. If $\rho(f, g) < \delta$, then $|g(x \pm k) - g(x)| \ge |f(x \pm k) - f(x)| - 2\delta$, so $\Delta g(x,k) \ge \Delta f(x,k) - (M-n)/2$ and $\|\Delta_k g\| \ge (M+n)/2 > n$.

(c) U_n is dense. Let $f \in \mathcal{C}$ and $\epsilon' > 0$. By uniform continuity choose a partition of I on whose subintervals f varies by less than $\epsilon'/2$; the piecewise linear p interpolating f at the partition points satisfies $\rho(f, p) < \epsilon'/2$ ($p(x)$ is a convex combination of two values each within $\epsilon'/2$ of $f(x)$). Let L bound the slopes of p, so $|p(x) - p(y)| \le L|x - y|$. Choose m with $r = m\epsilon'/4 \ge L + n + 1$, and let ϕ be the sawtooth that is linear on each $[j/m, (j+1)/m]$, alternately rising from 0 to $\epsilon'/4$ and falling back, so $0 \le \phi \le \epsilon'/4$ and ϕ has slope $\pm r$. Put $g = p + \phi$, so $\rho(f, g) < \epsilon'$, and $k = \min(1/n, 1/2m)$. Each x lies in some $[j/m, (j+1)/m]$ of length $1/m \ge 2k$, so $x + k$ or $x - k$ lies in the same subinterval,

where $|\phi(x \pm k) - \phi(x)| = rk$. Then $|g(x \pm k) - g(x)| \geq rk - Lk$, so $\Delta g(x,k) \geq r - L \geq n+1$ for all x, and $g \in U_n$.

By Lemma 8.1, $\bigcap U_n$ is dense, so it contains some g with $\rho(h,g) < \epsilon$. ■

8.3 Introduction to dimension theory

Motivation. Covering dimension m means every open cover has a refinement of order $m+1$; every compact m-manifold embeds in $\mathbb{R}^{2m+1}$.

A collection $\mathcal{A}$ has **order** $m+1$ if some point lies in $m+1$ elements and no point lies in more than $m+1$. X is **finite dimensional** if for some m every open covering has an open refinement covering X of order at most $m+1$; the least such m is the **topological dimension** $\dim X$. It is a topological invariant. Every compact subspace of $\mathbb{R}$ has dimension at most 1; an arc has dimension 1.

Theorem 8.7 (50.1)

If X is finite dimensional and $Y \subset X$ is closed, then Y is finite dimensional and $\dim Y \leq \dim X$.

Proof. Let $m = \dim X$ and $\mathcal{A}$ a covering of Y by sets open in Y; choose A' open in X with $A' \cap Y = A$. Cover X by the A' and $X - Y$, take an open refinement $\mathcal{B}$ of order at most $m+1$ covering X, and intersect its elements with Y: this refines $\mathcal{A}$, covers Y, and has order at most $m+1$. ■

Theorem 8.8 (50.2)

If $X = Y \cup Z$ with Y, Z closed and finite dimensional, then $\dim X = \max\{\dim Y, \dim Z\}$.

Proof. Let $m = \max\{\dim Y, \dim Z\}$; $\dim X \geq m$ by Theorem 8.7.

Step 1. every open covering $\mathcal{A}$ of X has an open refinement covering X in which each point of Y lies in at most $m+1$ elements. Take a refinement $\mathcal{C}$ of $\{A \cap Y\}$ by sets open in Y, covering Y, of order at most $m+1$. For $C \in \mathcal{C}$ choose U_C open in X with $U_C \cap Y = C$ and $A_C \in \mathcal{A}$ with $C \subset A_C$. The collection $\{U_C \cap A_C\} \cup \{A - Y : A \in \mathcal{A}\}$ works: it covers X, refines $\mathcal{A}$, is open since Y is closed, and a point $y \in Y$ in $U_C \cap A_C$ lies in C.

Step 2. given $\mathcal{A}$, apply Step 1 to get $\mathcal{B}$ (good at points of Y), then Step 1 for Z to $\mathcal{B}$ to get $\mathcal{D}$ (good at points of Z). Choose $f : \mathcal{D} \to \mathcal{B}$ with $D \subset f(D)$ and let $E(B) = \bigcup\{D : f(D) = B\}$. The sets $E(B)$ are open, cover X and refine $\mathcal{A}$. If x lies in distinct $E(B_1), \ldots, E(B_k)$, then x lies in distinct $B_1, \ldots, B_k$ and in distinct $D_1, \ldots, D_k$ with $f(D_i) = B_i$; so $k \leq m+1$ whether $x \in Y$ or $x \in Z$. ■

Corollary 8.9 (50.3)

If $X = Y_1 \cup \cdots \cup Y_k$ with each Y_i closed and finite dimensional, then $\dim X = \max\{\dim Y_1, \ldots, \dim Y_k\}$.

Proof. Induction on k using Theorem 8.8. ■

Points $x_0, \ldots, x_k$ of $\mathbb{R}^N$ are **geometrically independent** if $\sum a_i x_i = 0$ and $\sum a_i = 0$ imply all $a_i = 0$; equivalently, no x_j lies in the plane P spanned by the others, $P = \{\sum t_i x_i : \sum t_i = 1\}$. A set of points of $\mathbb{R}^N$ is in **general position** if each subset of at most $N+1$ of its points is geometrically independent.

Lemma 8.10 (50.4)

Given $x_1, \dots, x_n \in \mathbb{R}^N$ and $\delta > 0$, there are $y_1, \dots, y_n$ in general position with $|x_i - y_i| < \delta$.

Proof. Induction: let $y_1 = x_1$, and suppose $y_1, \dots, y_p$ are in general position. Each subset T of at most N of them spans a plane $P(T)$ of dimension at most $N-1$, a closed set with empty interior. A finite union of closed sets with empty interior has empty interior (if $U \subset A \cup B$ is open and nonempty, then $U - A$ is a nonempty open subset of B). So choose y_{p+1} within δ of x_{p+1} outside every $P(T)$. A subset S of at most $N+1$ of $y_1, \dots, y_{p+1}$ is independent: either $S \subset \{y_1, \dots, y_p\}$, or $S = T \cup \{y_{p+1}\}$; then if $\sum_T a_i y_i + a y_{p+1} = 0$ with $\sum a_i + a = 0$ and $a \neq 0$, $y_{p+1} = \sum(-a_i/a) y_i \in P(T)$, a contradiction; so $a = 0$ and all $a_i = 0$. ∎

Theorem 8.11 (50.5, imbedding theorem)

Every compact metrizable space X of topological dimension m imbeds in $\mathbb{R}^{2m+1}$.

Proof. Let $N = 2m+1$, $|\cdot|$ the square metric on $\mathbb{R}^N$, and $\mathcal{C}(X, \mathbb{R}^N)$ with the sup metric ρ, complete and hence Baire. For continuous f let $\Delta(f) = \sup\{\operatorname{diam} f^{-1}(z) : z \in f(X)\}$ and $U_\epsilon = \{f : \Delta(f) < \epsilon\}$.

U_ϵ is open: let $\Delta(f) < b < \epsilon$. The set $A = \{(x, y) : d(x, y) \geq b\}$ is compact and $|f(x) - f(y)| > 0$ on it, with positive minimum δ (if $A \neq \emptyset$). If $\rho(f, g) < \delta/2$, then $g(x) \neq g(y)$ on A, so $\Delta(g) \leq b < \epsilon$.

U_ϵ is dense: let f and $\delta > 0$ be given. Cover X by open sets of diameter less than $\epsilon/2$ whose images under f have diameter less than $\delta/2$; by compactness and $\dim X = m$ there is a finite open refinement $\{U_1, \dots, U_n\}$ covering X of order at most $m+1$, and a partition of unity $\{\phi_i\}$ dominated by it (Theorem 4.16). Choose $x_i \in U_i$ and, by Lemma 8.10, points z_i in general position with $|z_i - f(x_i)| < \delta/2$. Let $g = \sum \phi_i z_i$. If $\phi_i(x) > 0$, then $x \in U_i$ and $|f(x) - z_i| < \delta$; since $f(x) - g(x) = \sum \phi_i(x)(f(x) - z_i)$, $|f(x) - g(x)| < \delta$, and by compactness $\rho(f, g) < \delta$. If $g(x) = g(y)$, then $\sum(\phi_i(x) - \phi_i(y)) z_i = 0$ with coefficients summing to 0, and only indices with $x \in U_i$ or $y \in U_i$ appear, at most $2(m+1) = N+1$ of them. By general position all coefficients vanish. Choosing i with $\phi_i(x) > 0$ gives $\phi_i(y) > 0$, so $x, y \in U_i$ and $d(x, y) < \epsilon/2$. Hence $\Delta(g) \leq \epsilon/2$ and $g \in U_\epsilon$.

By Lemma 8.1, $\bigcap_n U_{1/n}$ is dense, hence nonempty; a map g in it is injective, hence an imbedding (Theorem 3.20). ∎

Theorem 8.12 (50.6)

Every compact subspace of $\mathbb{R}^N$ has topological dimension at most N.

Proof.

Step 1. an open covering of $\mathbb{R}^N$ by sets of square-metric diameter at most 1, of order at most $N+1$. The cubical cells of $\mathbb{R}^N$ are the products $\sigma = \prod I_i$ with each I_i either $\{k_i\}$ ($k_i \in \mathbb{Z}$) or $(k_i, k_i + 1)$; let $P(\sigma)$ be the set of indices of the first kind. Every point lies in exactly one cell. Write $t_i(y)$ for the distance from y_i to $\mathbb{Z}$, and let

$$V(\sigma) = \Big\{y : \ |y_i - k_i| < \tfrac{1}{2} \ (i \in P(\sigma)), \quad y_i \in (k_i, k_i + 1) \ (i \notin P(\sigma)), \quad \max_{i \in P(\sigma)} t_i(y) < \min_{i \notin P(\sigma)} t_i(y)\Big\},$$

an open set of diameter at most 1 containing σ. If $\sigma \neq \tau$ and $|P(\sigma)| = |P(\tau)|$, then $V(\sigma) \cap V(\tau) = \emptyset$: if $P(\sigma) = P(\tau)$ some k_i differs, which is impossible on a point coordinate (nearest integer) and on an interval coordinate (disjoint intervals); otherwise there are $i \in P(\sigma) - P(\tau)$ and $j \in P(\tau) - P(\sigma)$, and a common point would satisfy $t_i < t_j$ and $t_j < t_i$. So each point lies in at most one $V(\sigma)$ for each value $|P(\sigma)| \in \{0, \dots, N\}$, hence in at most $N+1$ of them.

Step 2. let $X \subset \mathbb{R}^N$ be compact, and let $\mathcal{A}$ be an open covering of X with Lebesgue number δ (Lemma 3.28). Choose $0 < s < \delta$; the sets $sV(\sigma) \cap X$ have diameter less than δ, so each lies in an element of $\mathcal{A}$; they form an open refinement covering X of order at most $N+1$. ■

Corollary 8.13 (50.7)

Every compact m-manifold has topological dimension at most m.

Proof. Each x has an open U with a homeomorphism $h : U \to W \subset \mathbb{R}^m$ open. Choose a closed ball $\overline{B} \subset W$ about $h(x)$; then $C_x = h^{-1}(\overline{B})$ is compact (so closed in X), contains the neighbourhood $h^{-1}(B)$ of x, and has dimension at most m by Theorem 8.12. Finitely many C_x cover X; apply Corollary 8.9. ■

Corollary 8.14 (50.8)

Every compact m-manifold imbeds in $\mathbb{R}^{2m+1}$.

Proof. It is compact Hausdorff, hence regular, with a countable basis, so metrizable (Theorem 4.12); its dimension k is at most m, and it imbeds in $\mathbb{R}^{2k+1} \subset \mathbb{R}^{2m+1}$ by Theorem 8.11. ■

Corollary 8.15 (50.9)

A compact metrizable space imbeds in some $\mathbb{R}^N$ iff it has finite topological dimension.

Proof. ($\Rightarrow$) Theorem 8.12.

($\Leftarrow$) Theorem 8.11. ■

Without compactness: every m-manifold, and more generally every metrizable space of dimension m with a countable basis, imbeds in $\mathbb{R}^{2m+1}$ (Menger–Nöbeling).

8.4 Selected exercises

Motivation. Exercises from Munkres on Baire's theorem in locally compact spaces, continuity points of functions on $\mathbb{R}$, and dimension and connectedness.

Exercise 8.1 (§48, Ex. 3; locally compact spaces are Baire)

Show that every locally compact Hausdorff space X is a Baire space.

Solution. Let $A_1, A_2, \ldots$ be closed with empty interior and U_0 nonempty open. We show $U_0 \not\subset \bigcup_n A_n$.

Step 1. Since A_1 has empty interior, $U_0 - A_1$ is a nonempty open set. Pick a point in it; by Theorem 3.35 it has a neighbourhood U_1 with $\overline{U_1}$ compact and $\overline{U_1} \subset U_0 - A_1$. Inductively, given a nonempty open U_{n-1}, choose a nonempty open U_n with $\overline{U_n}$ compact and $\overline{U_n} \subset U_{n-1} - A_n$.

Step 2. The sets $\overline{U_n}$ are nonempty, closed in the compact space $\overline{U_1}$, and nested, so they have the finite intersection property and $\bigcap_n \overline{U_n} \neq \emptyset$ (Theorem 3.23).

Step 3. A point x of the intersection lies in $\overline{U_1} \subset U_0$, and for each n, $x \in \overline{U_n} \subset U_{n-1} - A_n$, so $x \notin A_n$. Hence $U_0 \not\subset \bigcup A_n$: the union has empty interior. ■

Exercise 8.2 (§48, Ex. 7; continuity exactly on a countable dense set)

Let D be a countable dense subset of $\mathbb{R}$. Show that there is no $f : \mathbb{R} \to \mathbb{R}$ that is continuous precisely at the points of D.
(a) Show that the set C of points at which a function $f : \mathbb{R} \to \mathbb{R}$ is continuous is a G_δ set.
(b) Show that D is not a G_δ set.

Solution.

(a) For $U \subset \mathbb{R}$ let $\operatorname{diam} f(U) = \sup\{|f(s) - f(t)| : s, t \in U\}$, and let U_n be the union of all open U with $\operatorname{diam} f(U) < 1/n$; U_n is open.

Step 1. $C \subset \bigcap U_n$: if f is continuous at x, choose $\delta > 0$ with $|f(y) - f(x)| < 1/(3n)$ for $|y - x| < \delta$. Then $U = (x - \delta, x + \delta)$ has $\operatorname{diam} f(U) \le 2/(3n) < 1/n$, so $x \in U_n$.

Step 2. $\bigcap U_n \subset C$: let $x \in \bigcap U_n$ and $\epsilon > 0$; choose n with $1/n < \epsilon$. Then x lies in an open U with $\operatorname{diam} f(U) < 1/n$, so $|f(y) - f(x)| < \epsilon$ for all $y \in U$. Thus f is continuous at x, and $C = \bigcap_n U_n$.

(b) Suppose $D = \bigcap_n W_n$ with W_n open. Each $W_n \supset D$ is dense, and for $d \in D$ the open set $V_d = \mathbb{R} - \{d\}$ is dense. The family $\{W_n\} \cup \{V_d : d \in D\}$ is countable, and

$$\bigcap_n W_n \cap \bigcap_{d \in D} V_d = D - D = \emptyset.$$

But $\mathbb{R}$ is a complete metric space, hence a Baire space (Theorem 8.2), so this countable intersection of dense open sets is dense by Lemma 8.1, a contradiction.

By (a) and (b), the set of continuity points of f is never D. ■

Exercise 8.3 (§50, Ex. 2; connected spaces have dimension at least 1)

Show that a connected T_1 space X with more than one point has dimension at least 1.

Solution. If X is not finite dimensional there is nothing to prove, so suppose $\dim X = 0$ and derive a contradiction.

Step 1. Choose $x \neq y$. As X is T_1, $\mathcal{A} = \{X - \{x\}, X - \{y\}\}$ is an open covering. Since $\dim X = 0$, it has an open refinement $\mathcal{B}$ covering X of order at most 1, i.e. the elements of $\mathcal{B}$ are pairwise disjoint.

Step 2. Let $V \in \mathcal{B}$ contain x. V lies in an element of $\mathcal{A}$, which must be $X - \{y\}$ since $x \in V$; so $y \notin V$. Let W be the union of the other elements of $\mathcal{B}$.

Step 3. V and W are open, disjoint, cover X, and are nonempty ($x \in V$, $y \in W$). This is a separation of the connected space X, a contradiction. ■